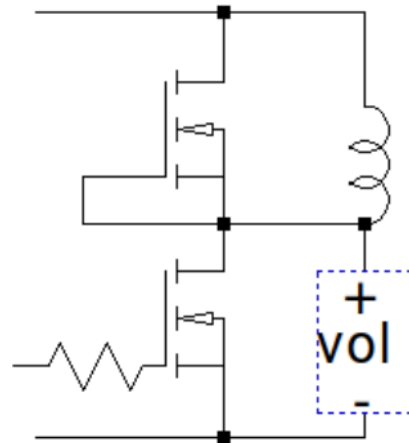
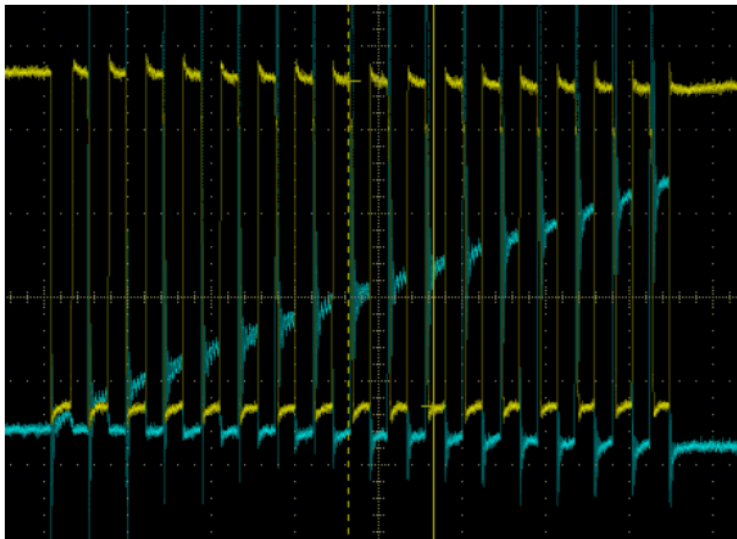


Gate driver and double pulse tests



- CH3 : OUTA
- CH4 : OUTB

Driver:

ucc21520Q

Programmed dead time

RT = 11.5k, so dead time = 115ns

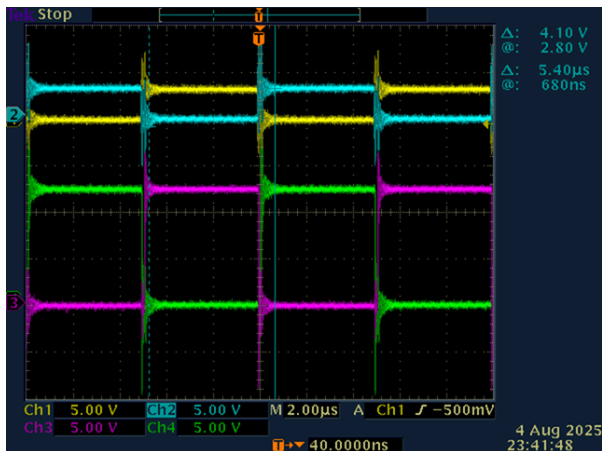


Figure 2 Zoom out to see 2 periods, 100khz

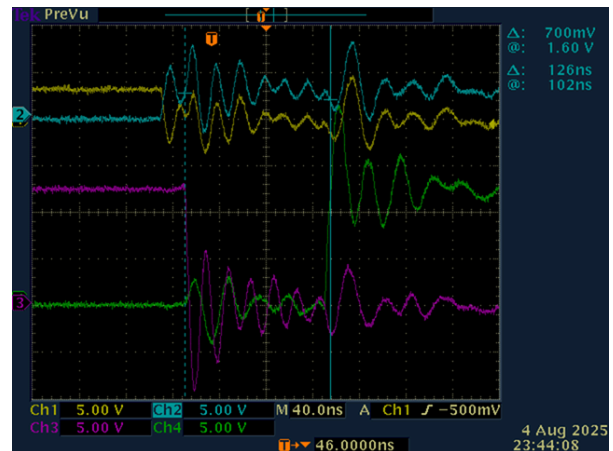


Figure 3 Zoom in around the dead time:
DT=126ns

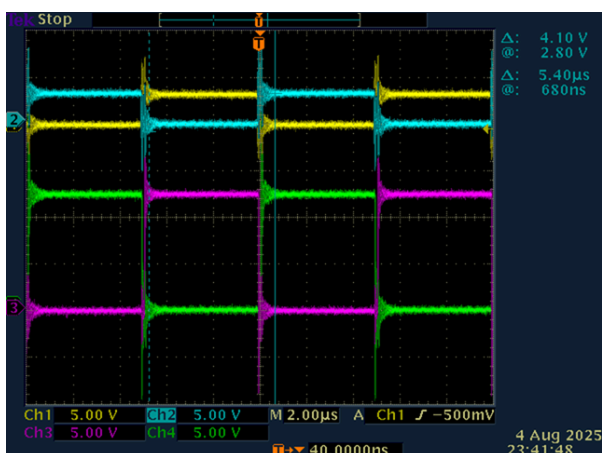


Figure 4 Zoom out to see 2 periods, 100khz

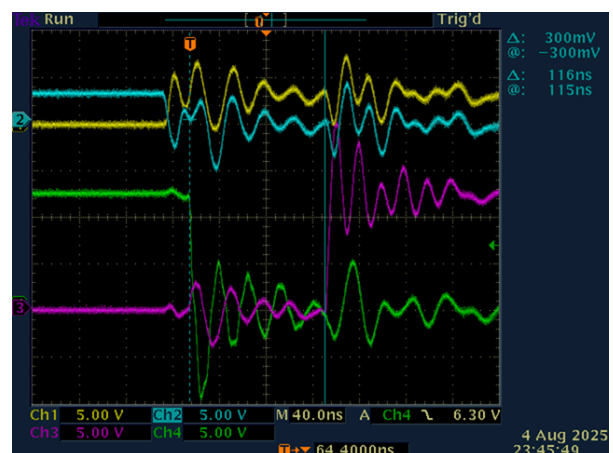


Figure 5 Zoom in around the dead time:
DT=116ns

$$\begin{aligned} \mathrm{DT}_{\mathrm{Prog}} &= 115 \text{ ns} \\ \mathrm{DT}_{\mathrm{Meas}} &= 116 \text{ ns} \\ \mathrm{Error} &= 100 \cdot \frac{\mathrm{DT}_{\mathrm{Meas}} - \mathrm{DT}_{\mathrm{Prog}}}{\mathrm{DT}_{\mathrm{Prog}}} = 100 \cdot \frac{116 - 115}{115} = 0.87 \% \end{aligned}$$

So the measured dead time is very accurate

Bootstrap circuit

Lest's check the bootstrap circuit (DRC)

SINGLE pulse

In this test, the low-side MOSFET is driven by a single long pulse.

Peak current estimation :

$$\begin{aligned} \mathrm{V}_{12} &= 12 \text{ V} \\ \mathrm{V}_d &= 0.700 \text{ V} \\ \mathrm{R}_{boot} &= 2.150 \text{ } \\ \mathrm{I}_{boot} &= \frac{\mathrm{V}_{12} - \mathrm{V}_d}{\mathrm{R}_{boot}} = \frac{12 - 0.700}{2.150} = 5.256 \end{aligned}$$

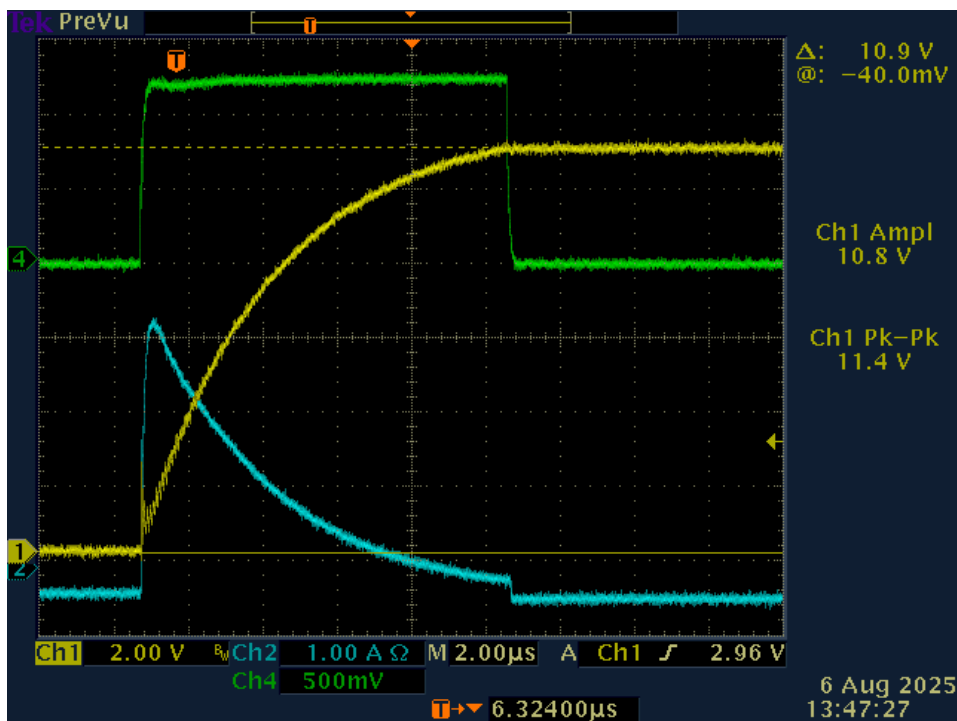


Figure 6 Bootstrap circuit measurement with 10us pulse width

Oscilloscope:

- **CH1** : bootstrap capacitor voltage
- **CH2** : bootstrap diode current
- **CH4** : Vgs low side MOSFET

Peak current measurement :

$$I_{bootMeasure} \sim 3.8 \text{ A}_{peak}$$

The measured peak current is lower than the estimated value because the estimate was based on the assumption that the only resistance in the path was the 2.15-Ω resistor, which represents a pessimistic (worst-case) assumption.

Below is a fast/simplified reproduction of this test in simulation

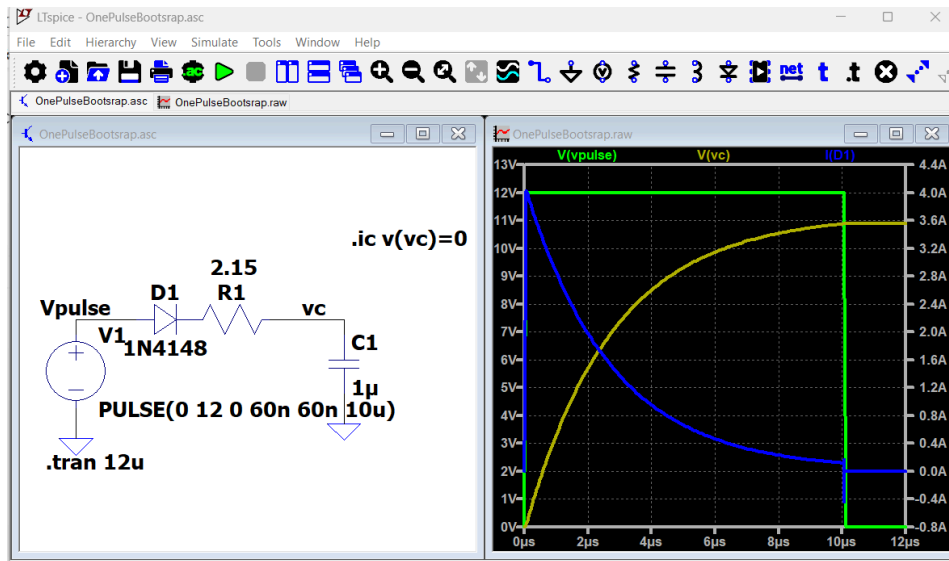


Figure 7 Quick simulation of the bootstrap circuit with 10us pulse width

PWM mode

In this test, the low-side MOSFET is driven by a pulse train with frequency of 100kHz (more realistic).

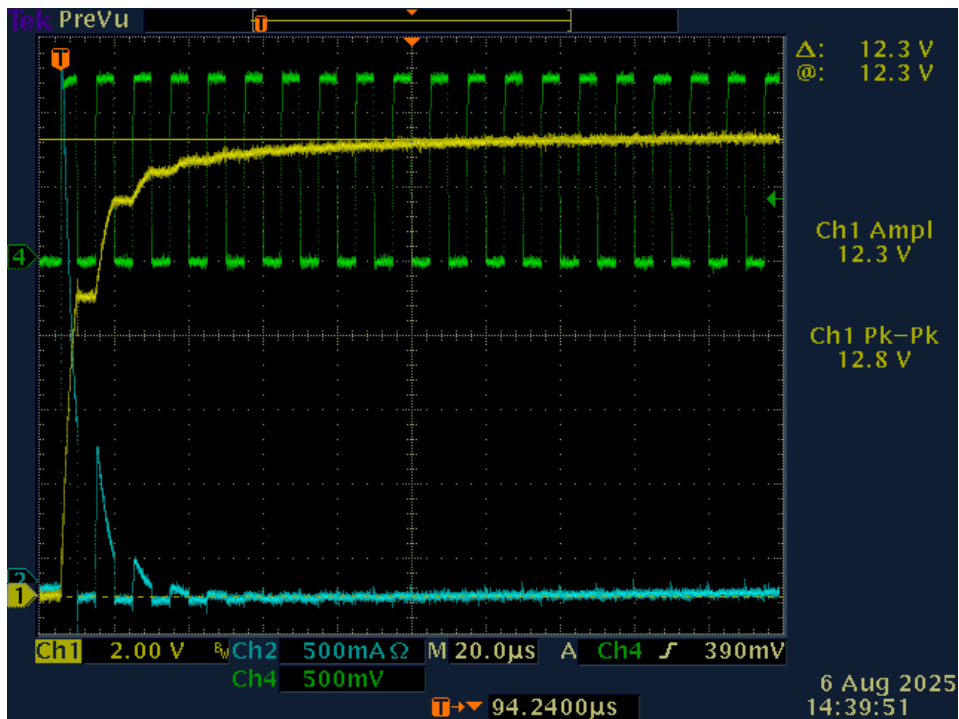


Figure 8 Bootstrap circuit measurement with PWM 100kHz 50% duty cycle

Below is a fast/simplified reproduction of this test in simulation

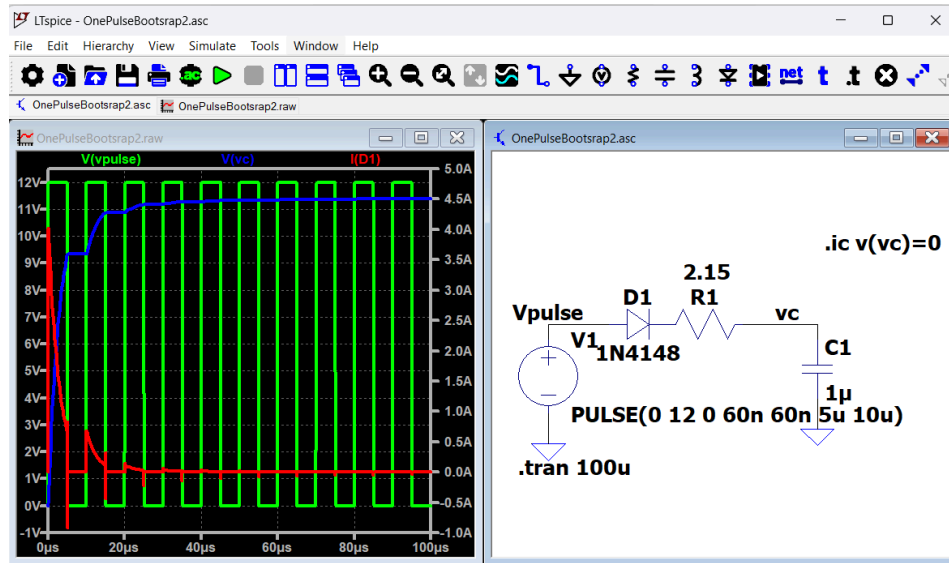


Figure 9 Quick simulation of the bootstrap circuit 100kHz 50% duty cycle

Estimation of the charge time of the bootstrap capacitor:

$$\begin{aligned} C &= 1 \text{ } \mu\text{s} \\ R &= 2.150 \text{ } \Omega \\ \tau &= R \cdot C = 2.150 \cdot 1 = 2.150 \text{ } \mu\text{s} \end{aligned}$$

3 corresponds to the 99% level, and 2 represents the 50% duty cycle.

$$\begin{aligned} \text{time} &= \tau \cdot 5 \cdot 2 = 2.150 \cdot 5 \cdot 2 = 21.500 \text{ } \mu\text{s} \end{aligned}$$

So minimum time to charge the capacitor is 22 μ s or 3 periods of 100khz

Double pulse test

Setup

We used the primary side of the DC-DC (without ferrite elements) to make the double pulse test.

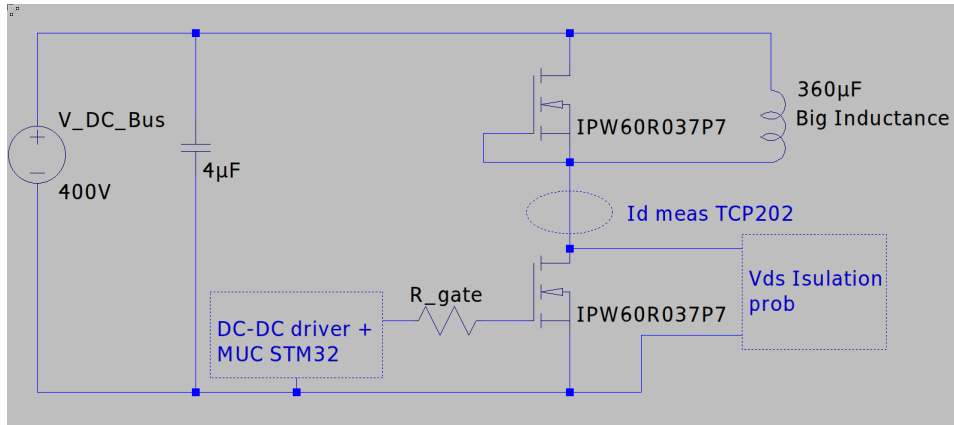


Figure 10 Double pulse setup

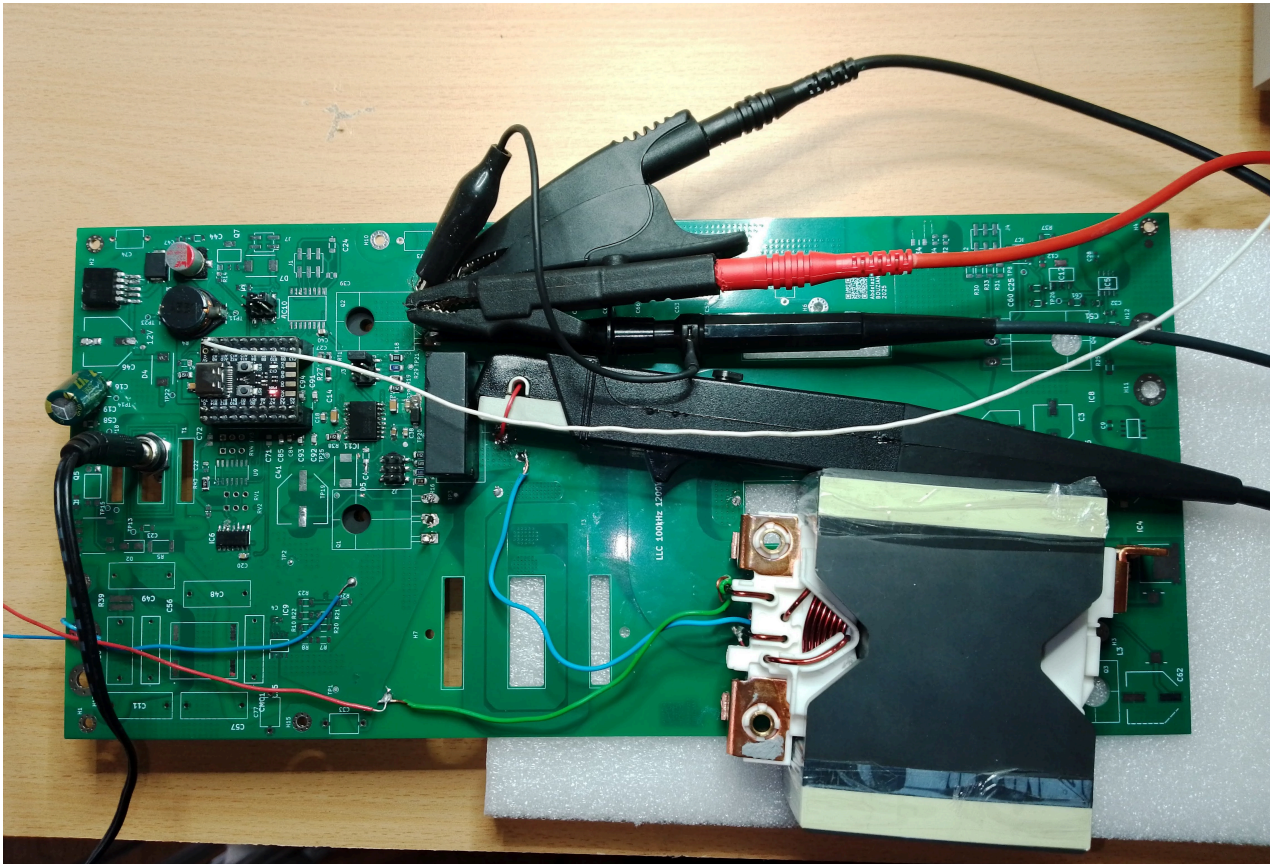


Figure 11 Double pulse setup (with the big inductance 360μH, 16 turns)

Example of the pre-processing

In this section an example of pre-processing measured signal to calculate the E_{on} , and E_{off} energy

- $V_{bus} = 366.5V$
- $V_{lv} = 12.34v$
- $R_{gate_ext} = 10\Omega$

Step 1: Load the csv files

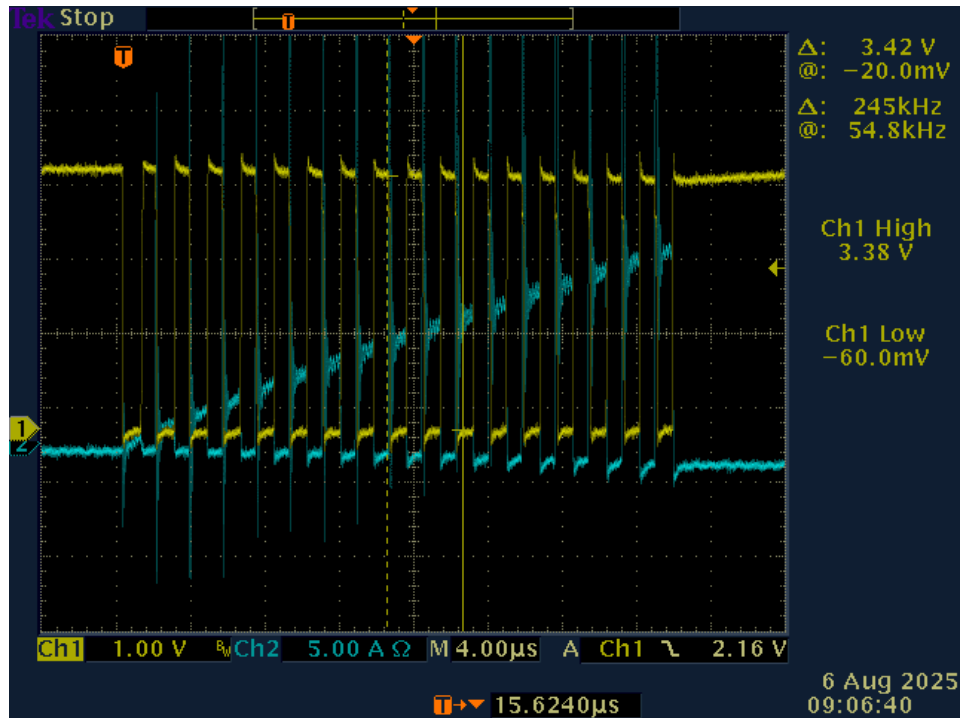
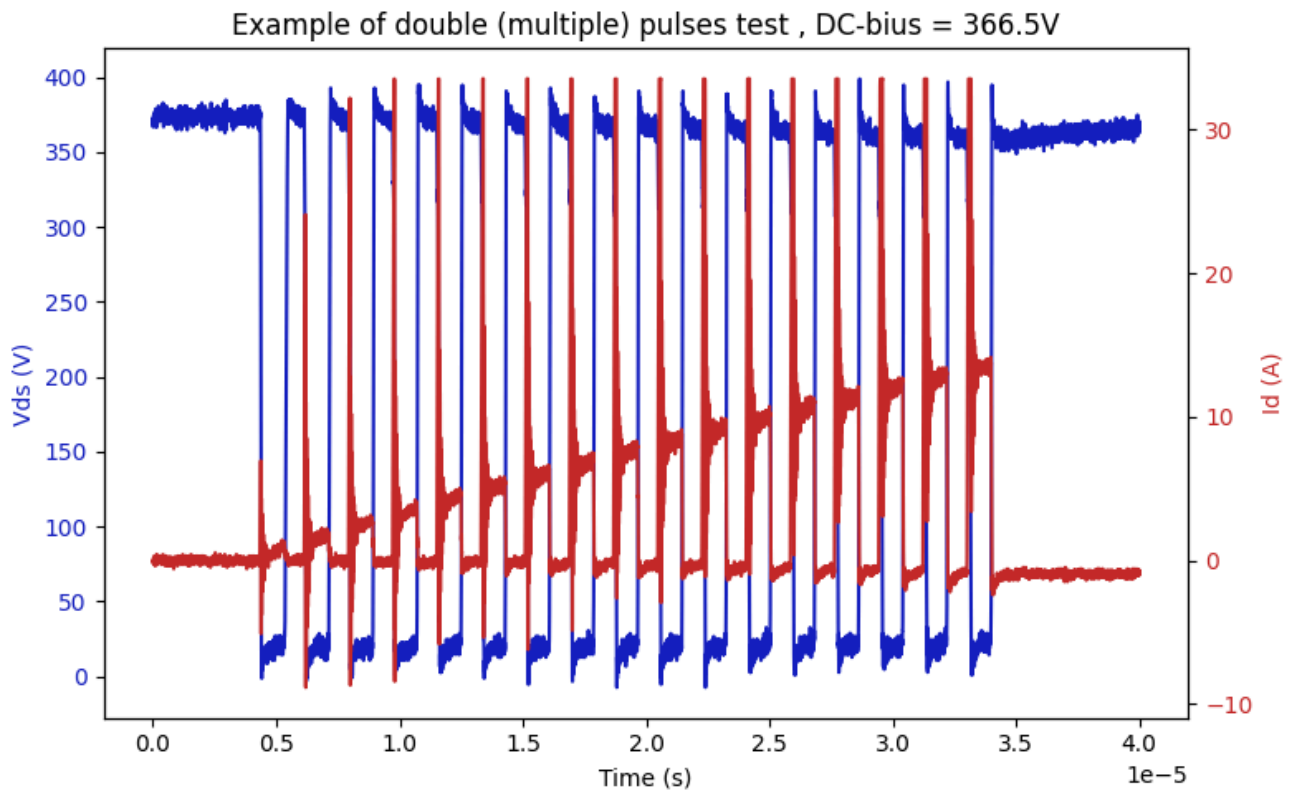


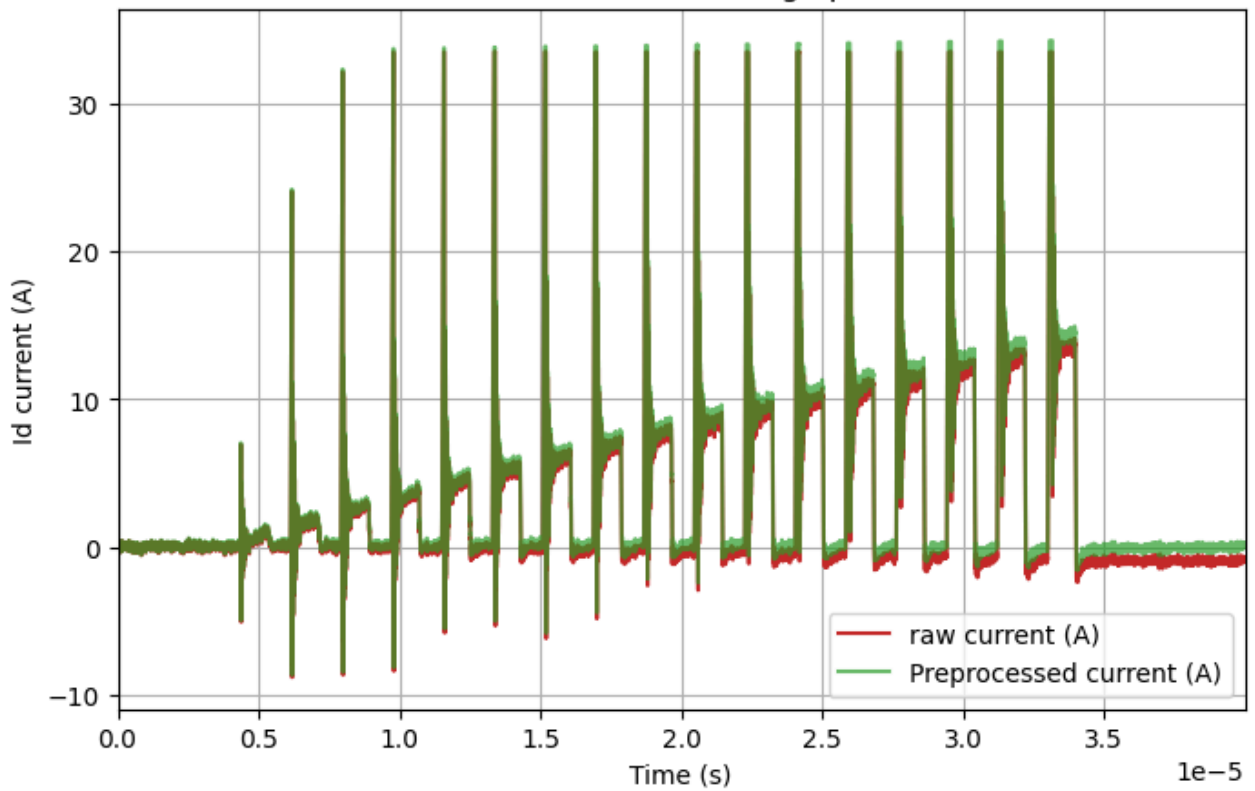
Figure 12 Quick simulation of the bootstrap circuit with 10us pulse width



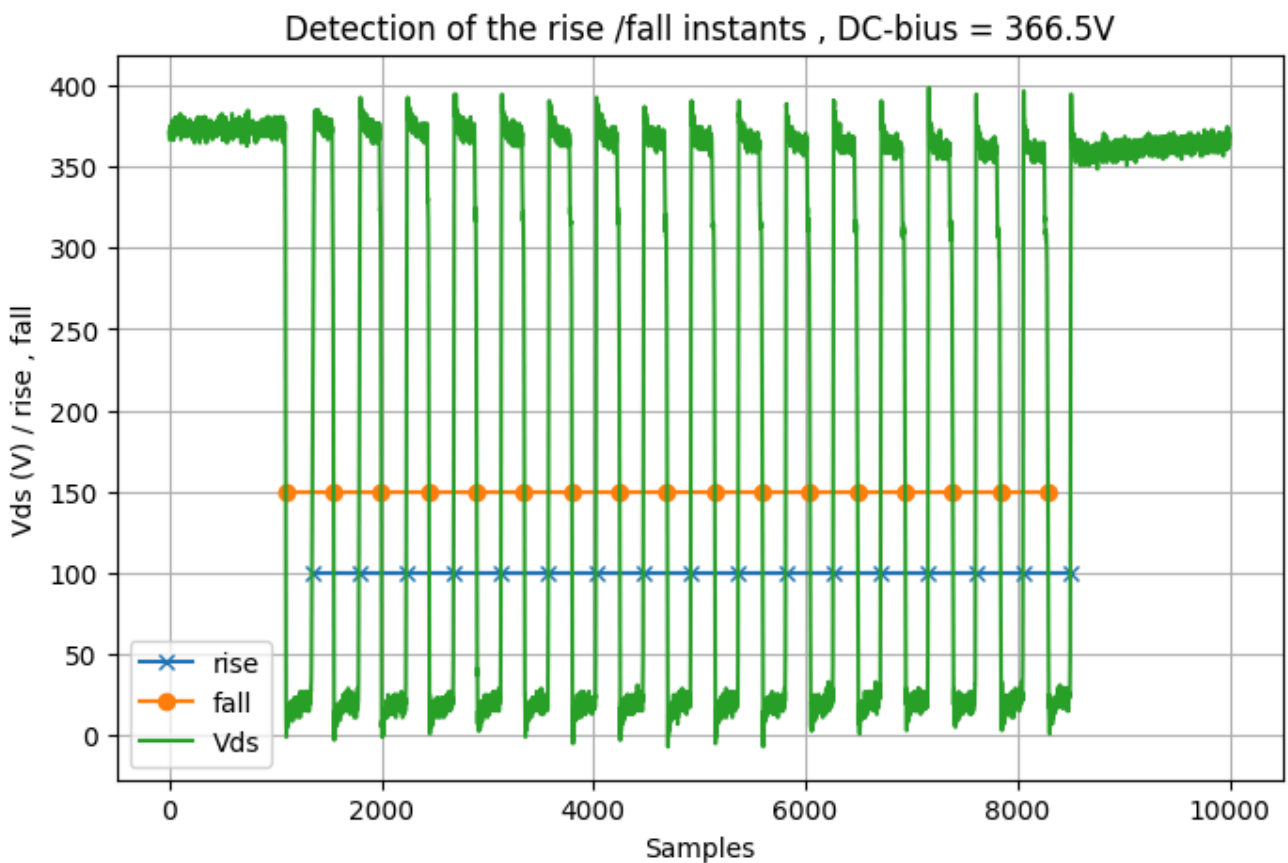
Step 2 : Pre-processing the current

The used current prob has a heigh pass filter effect, so below a pre-processing to remove this effect.

Correction of the current of ID: remove the heigh pass filter effect of the sensor

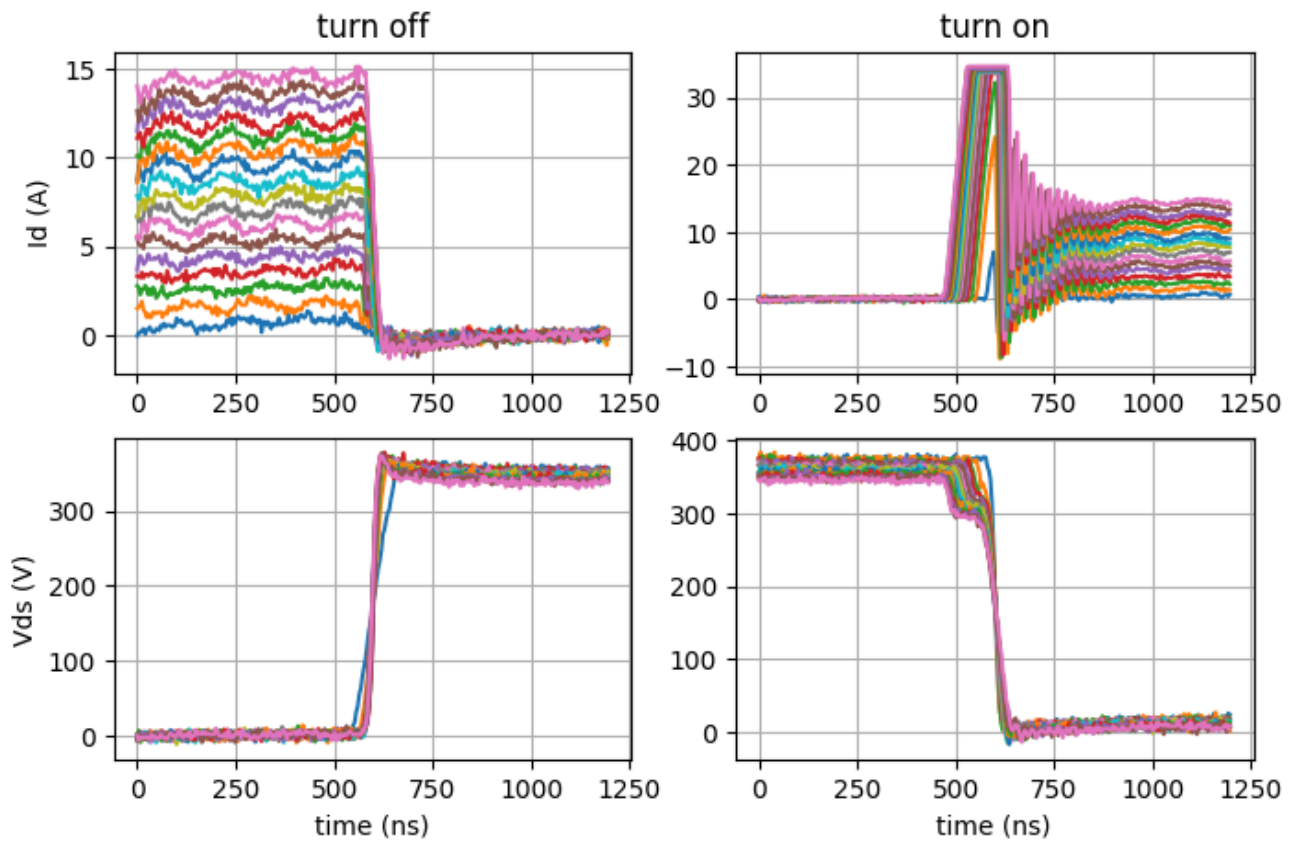


Step 3: Detection of the rise/fall instants

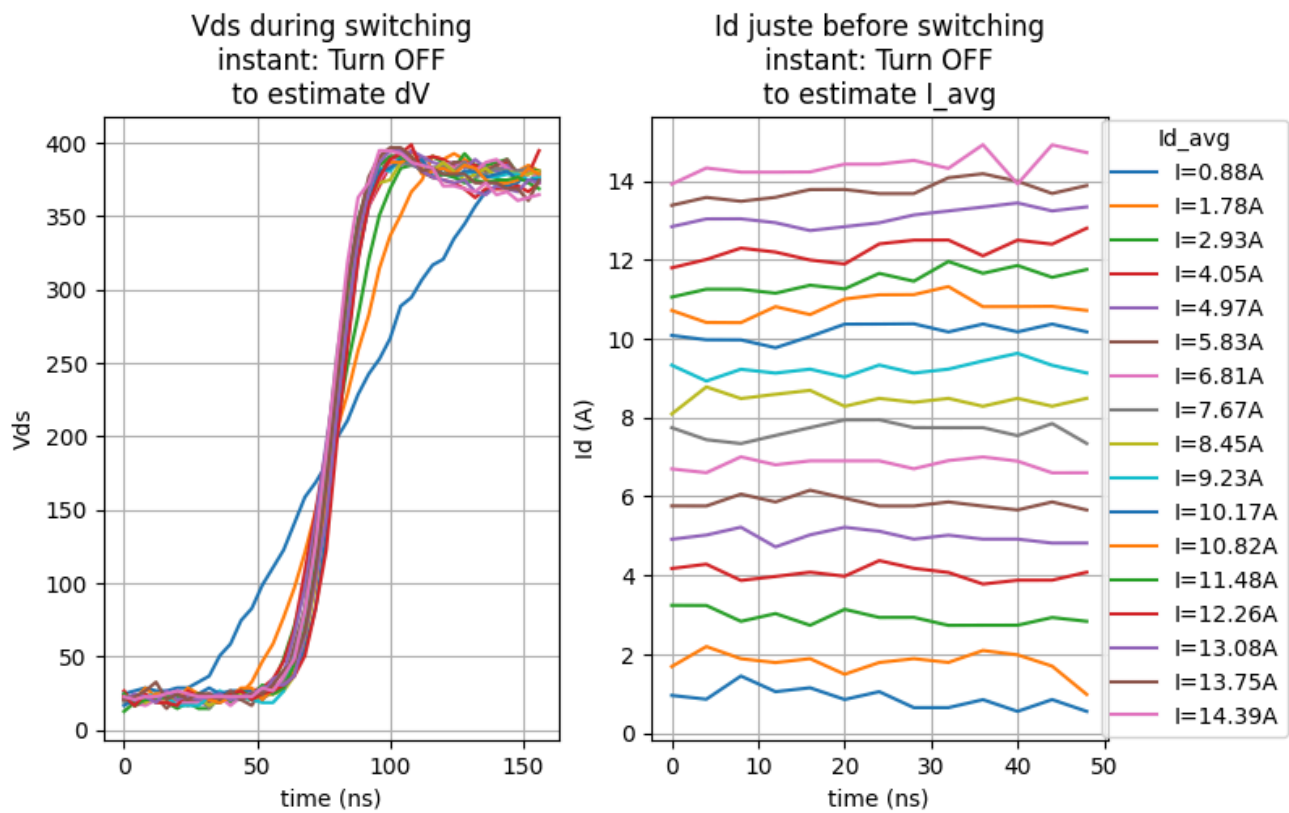


Step 4: zoom around the switching instants

Pre-Processed data of turn off/on instants, DC-bias = 366.5V



Step 5: Estimation of the Coss capacitor



Formulas to calculate the mosfet capacitors:

Time related capacitance

$$i = C \frac{dv}{dt}$$

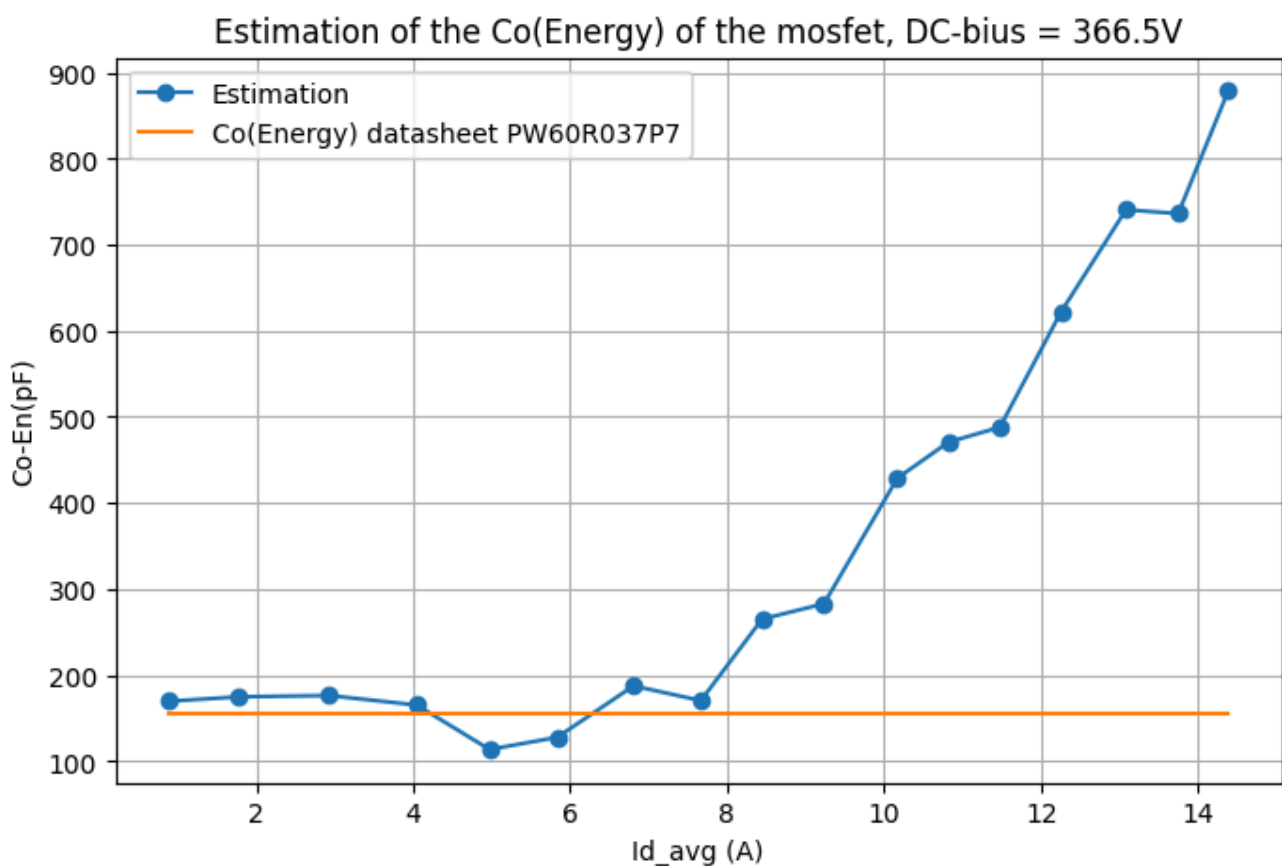
$$C_{o(tr)} = \frac{\int i \cdot dt}{\int dv} = \frac{Q}{\Delta V}$$

Energy-related capacitance

$$E = \int v \cdot i \cdot dt$$

$$C_{o(er)} = \frac{2E}{\Delta V^2}$$

dv is measured between $\min_v + 50V$ and $\max_v - 50V$



Why the estimation of Co-Energy is good at low current, and bad in high current: $E_{total} = E_{oss} + E_{switching_loss}$

In low current loss energy is low so almost all measured energy is the stored energy in the mosfet capacitor, but in high current a part of energy is converted to heat.

To reproduce this phenomenon, I propose the simple simulation below:

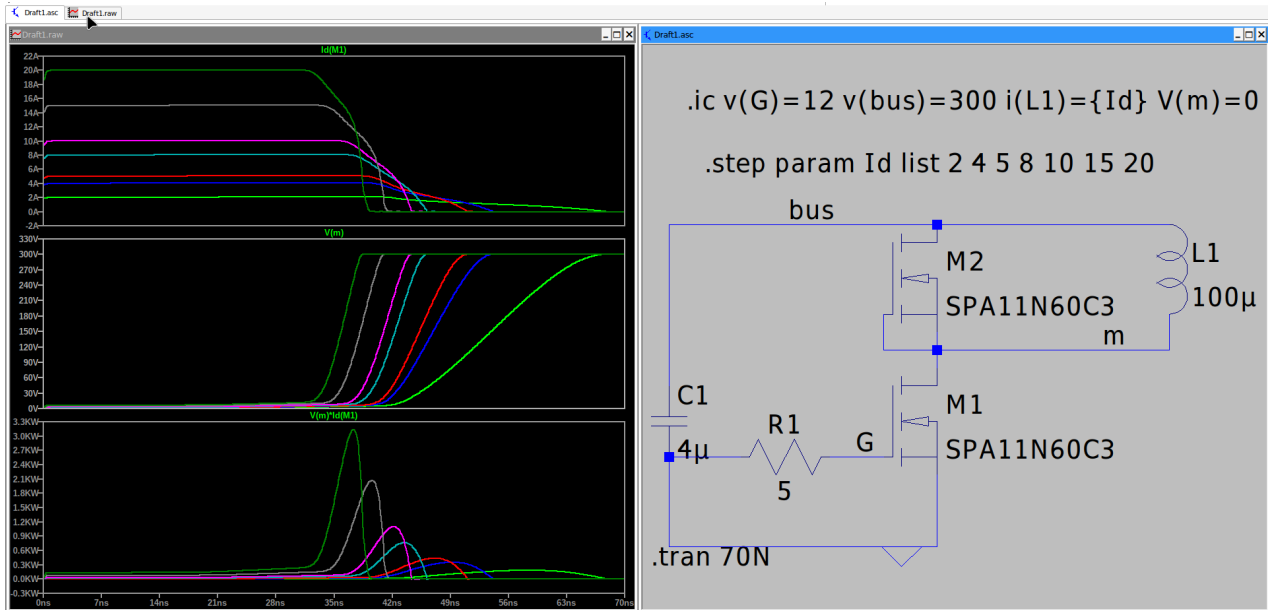
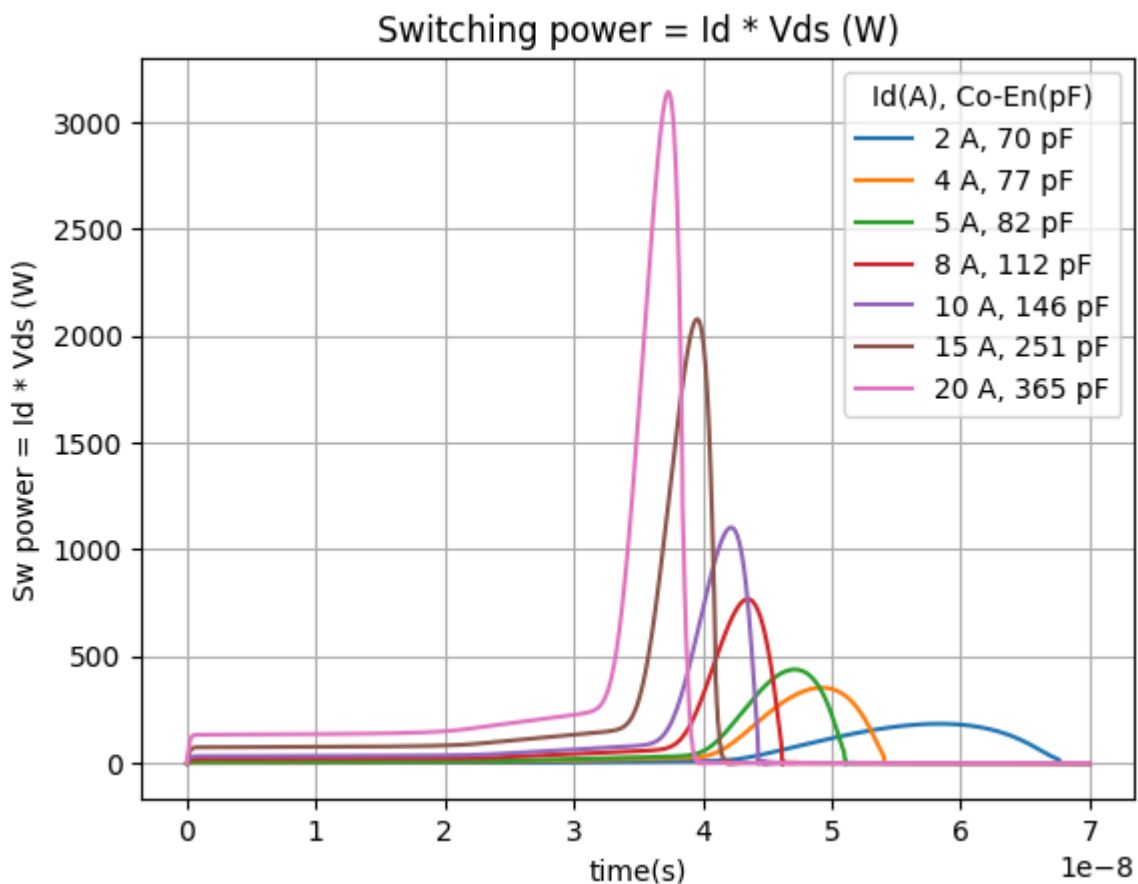
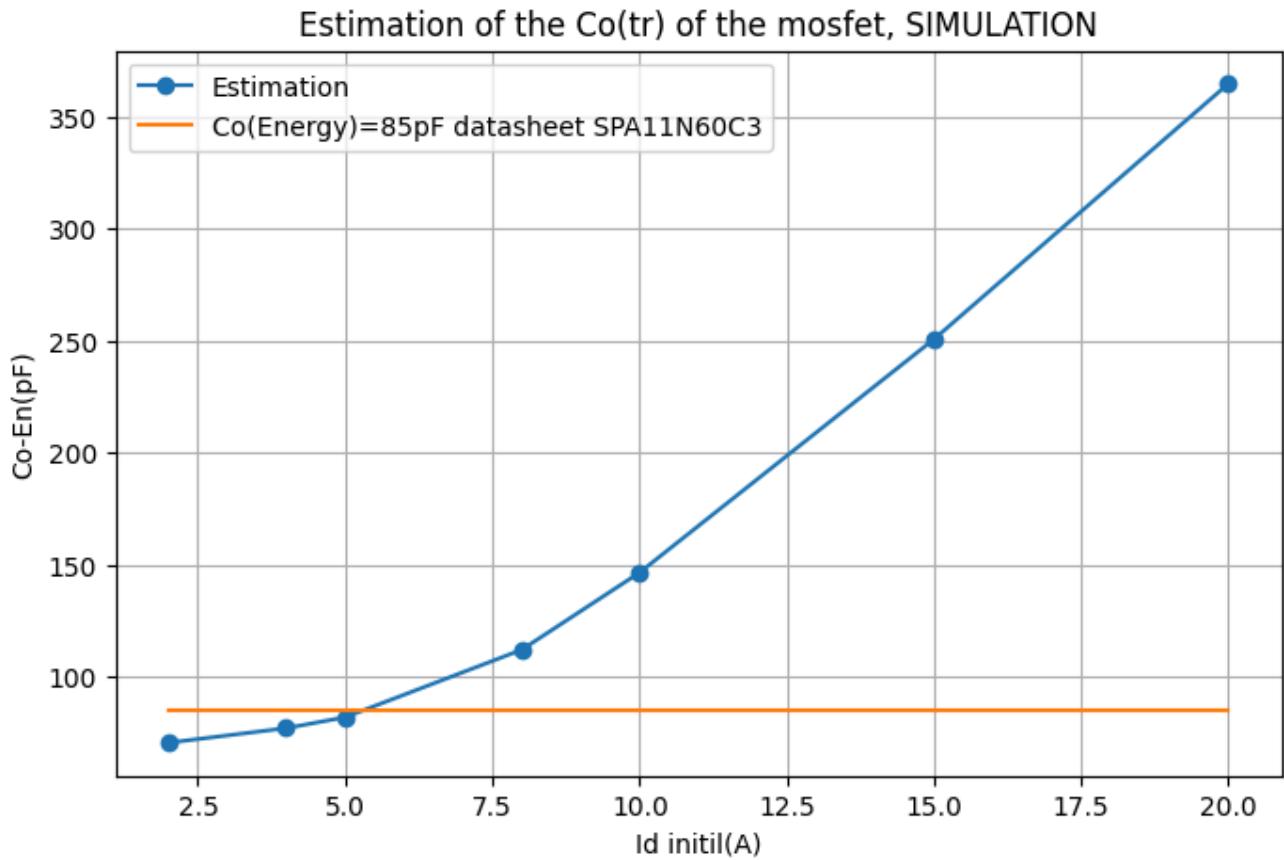


Figure 13 Simple simulation of the turning off instant of SPA11N60C3

Lest's calculating the sw-engry and the coresponding capacitor using pre-processing of the raw data using python

```
Text(0.5, 1.0, 'Switching power = Id * Vds (W)')
```

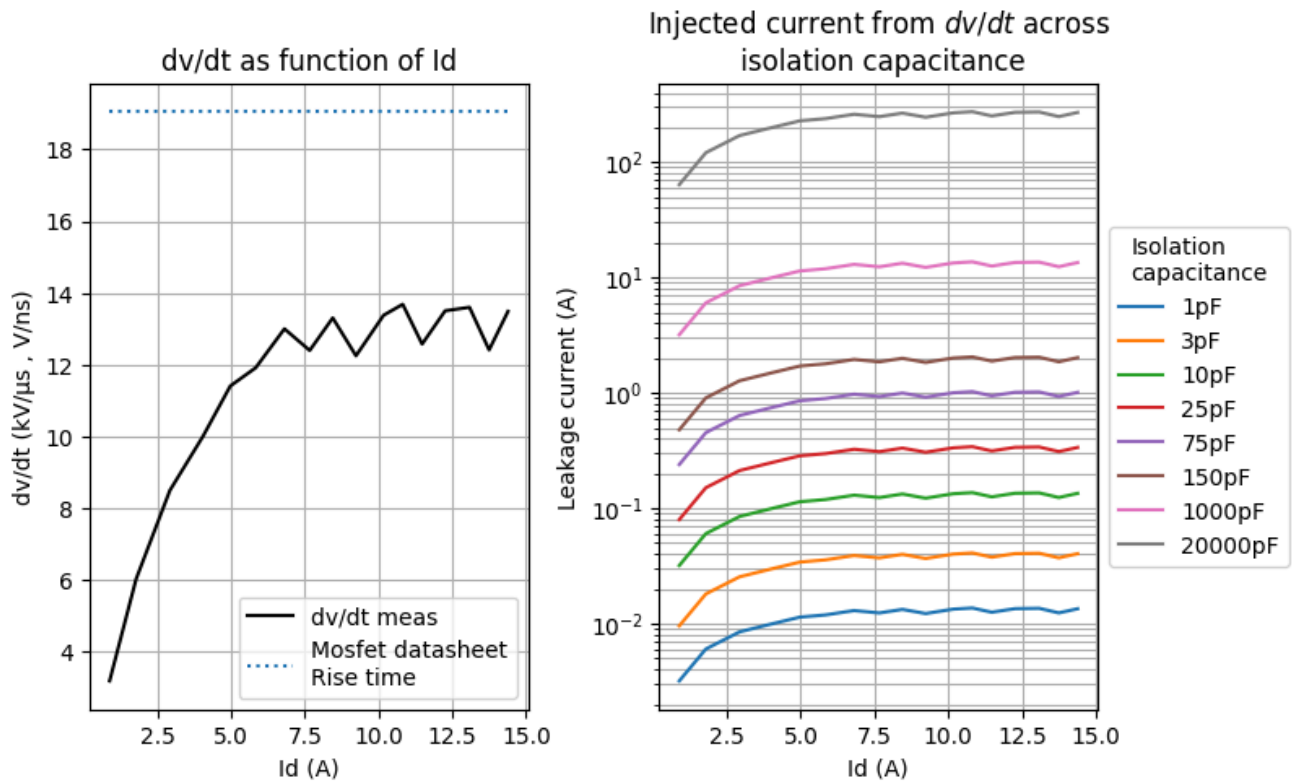




So , we can see the same behaviour our measurement

For information dv/dt:

Shown below are the dv/dt measured during turn-off and the corresponding current as a function of isolation capacitance, for the case where an isolated power supply is used to power the high-side driver.



In **IPW60R037P7** [datasheet](#), we can find :

- Rise time (t_r) = 21ns
- @ $V_{DD}=400V, V_{GS}=13V, I_D=29.5A, R_G=3.3\Omega$

So $\frac{dv}{dt} = 19V/ns$

The "dv/dt meas" has a maximum, this is explained by $\frac{dv}{dt}_{max} = \frac{V_{gsPlateau} - V_{th}}{R_{g_tot} \cdot C_{gd}}$

Since the driver is powered by 12V/0V (no V_{th}), so: $\frac{dv}{dt}_{max} = \frac{V_{gsPlateau}}{R_{g_tot} \cdot C_{gd}}$

where

C_{gd} : Gate-Drain cap, C_{rss} in datasheet

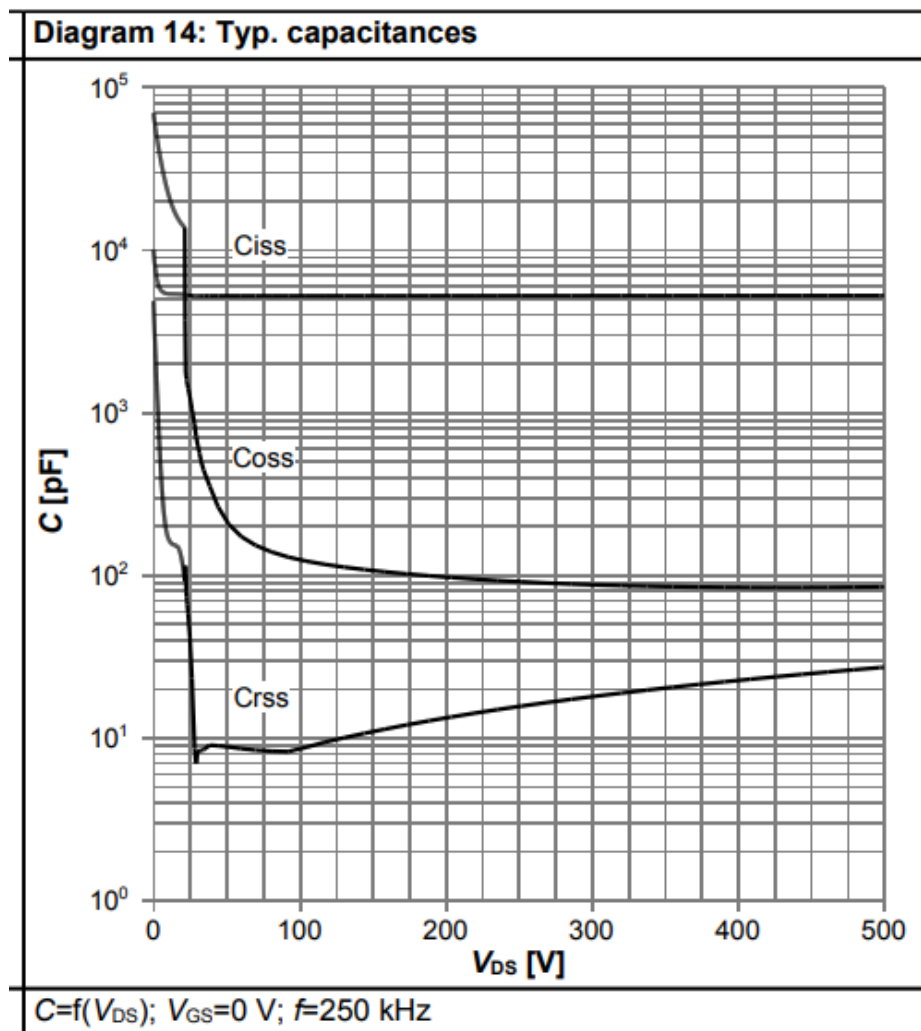


Figure 14 IPW60R037P7 datasheet: $C_{rss} = 22\text{ pF}$ at 366 V

Let's calculate the theoretical $dv/dt_{\max}$:

$$\begin{aligned} R_{\text{gMosfet}} &= 0.860 \, \Omega \, ; \, (\Omega, \text{ see IPW60R037P7 datasheet}) \, \, R_{\text{driver}} = 5 \, \Omega \, ; \\ & \, (\Omega, \text{ see UCC21520Q datasheet}) \, \, R_{\text{ext}} = 10 \, \Omega \, ; \, (\Omega, \text{ in this example}) \, \, R_{\text{gTot}} = \\ R_{\text{gMosfet}} + R_{\text{driver}} + R_{\text{ext}} &= 0.860 + 5 + 10 = 15.860 \, \Omega \end{aligned}$$

$$\begin{aligned} V_{\text{gsPlateau}} &= 5.200 \, \text{V} \, ; \, (\text{V, see IPW60R037P7 datasheet}) \, \, C_{\text{g}} = 22 \, \text{pF} \, ; \\ & \, (\text{pF, see fig above}) \, \, \text{d}v/\text{d}t = \frac{V_{\text{gsPlateau}}}{R_{\text{gTot}} \cdot C_{\text{g}} \cdot 1 \times 10^{-3}} \\ &= \frac{5.200}{15.860 \cdot 22 \cdot 1 \times 10^{-3}} = 14.903 \, \text{V/ns} \end{aligned}$$

The maximum dv/dt measured is around $13 - 14 \text{ V/ns}$, so our theoretical estimation is very close

Rg impact

In this section, we will change gate resistor, and check the impact in the loss energy.

We calculating the E_{on} just for information, since we are in ZVS, E_{on} is very low, and the calculated E_{on} is compatible with no ZVS function.

In the PCB we mount 3 resistors in parallel for R_g , 3 Ω , 5 Ω and 10 Ω with a jumper, so it will be easy to change R_g using this value, or any combinason of parallel configuration like 3 $\Omega // 5 \Omega = 1.875 \Omega$ and 3 $\Omega // 10 \Omega = 2.30 \Omega$ (see the photo below)

Just for information, we broked the mosfet during this test @400V, so we recommande to no make SW-ON.

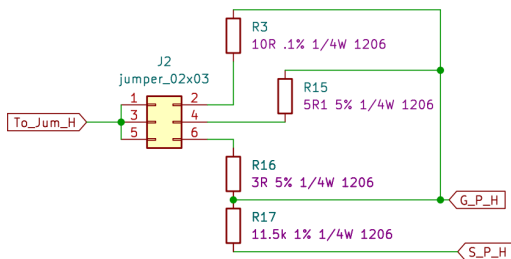


Figure 15 The Jumper used to select R_g

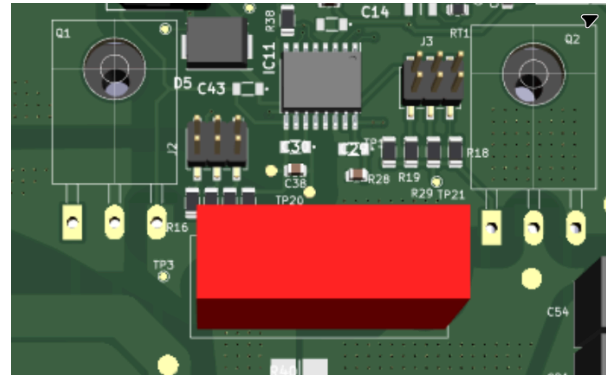
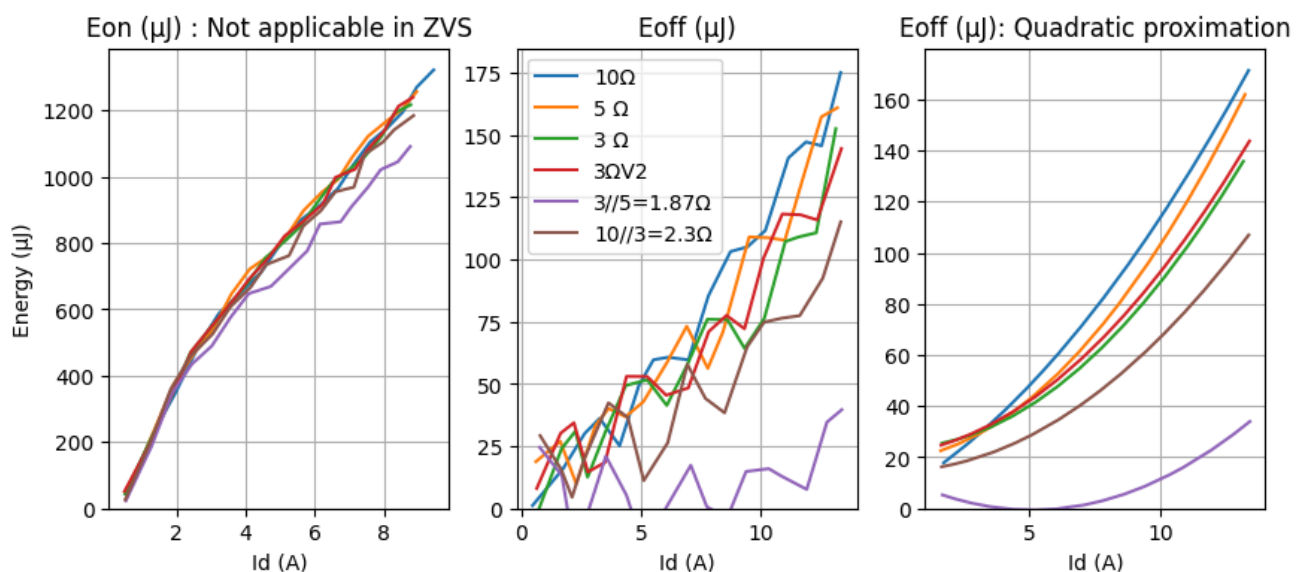


Figure 16 Jumper in the PCB with the 3 resostors (3D)

We followed the same steps as shown in the example above to calculate the E_{on}/E_{off} of each R_g configuration



$R_g = 3 \Omega$ is a good compromise between reducing the switching energy and avoiding oscillation, we will conserve this value for this project

Gate current

In this test, we measured the gate current using TCP202, so a small loop wire is added to the current path to insert the current probe (inductance)

The test was done in low voltage condition:

- $V_{in} = 48.03V$
- $V_{out} = 8.03V$
- Freq = 150kHz (Max freq of the DC-DC)

This test is done with the old primary mosfet (before updating the BOM)

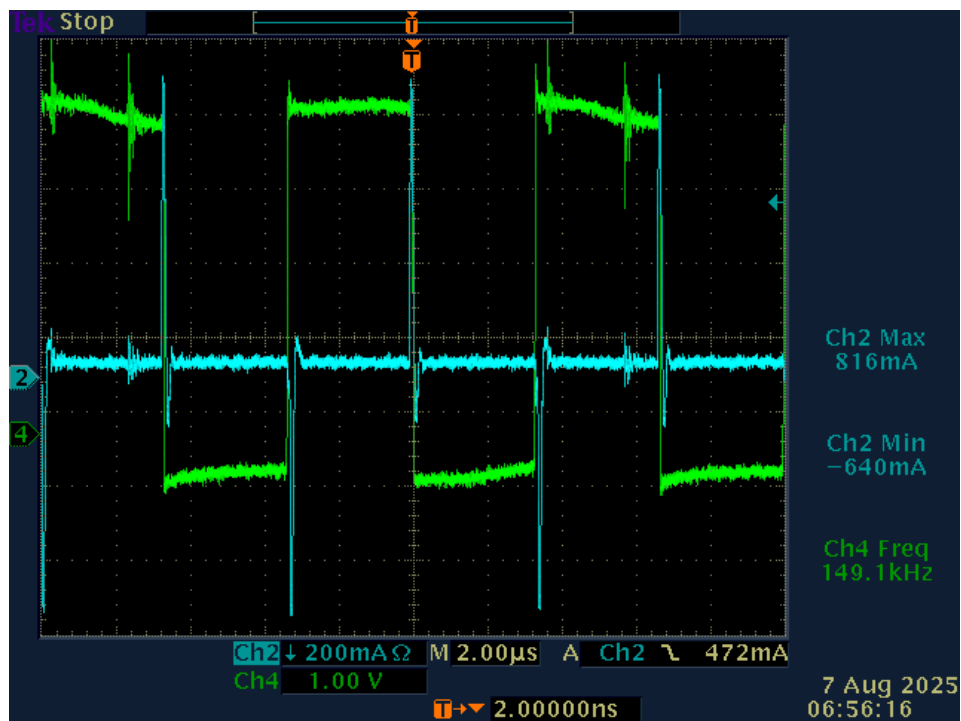
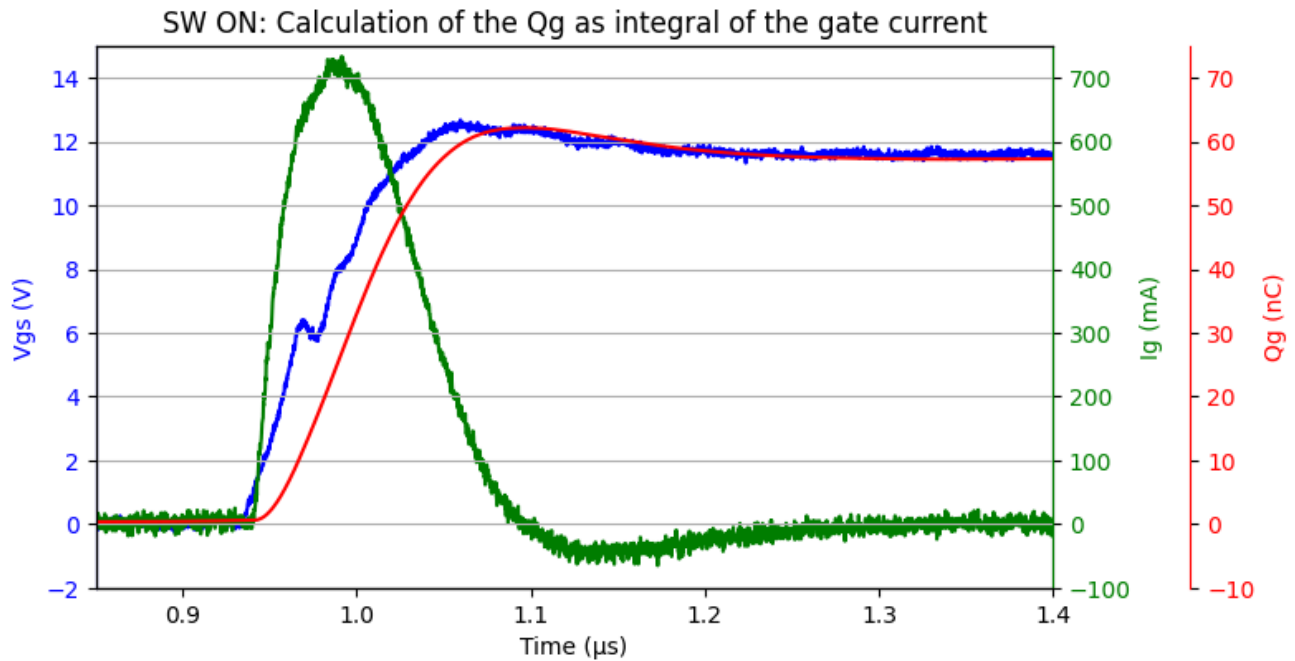


Figure 17 Primary-Low side gate current/voltage

Pre-Processing of the oscilloscope raw data



We can estimate the R_{gTot}

Let's consider that so inductance is present in the gate loop.

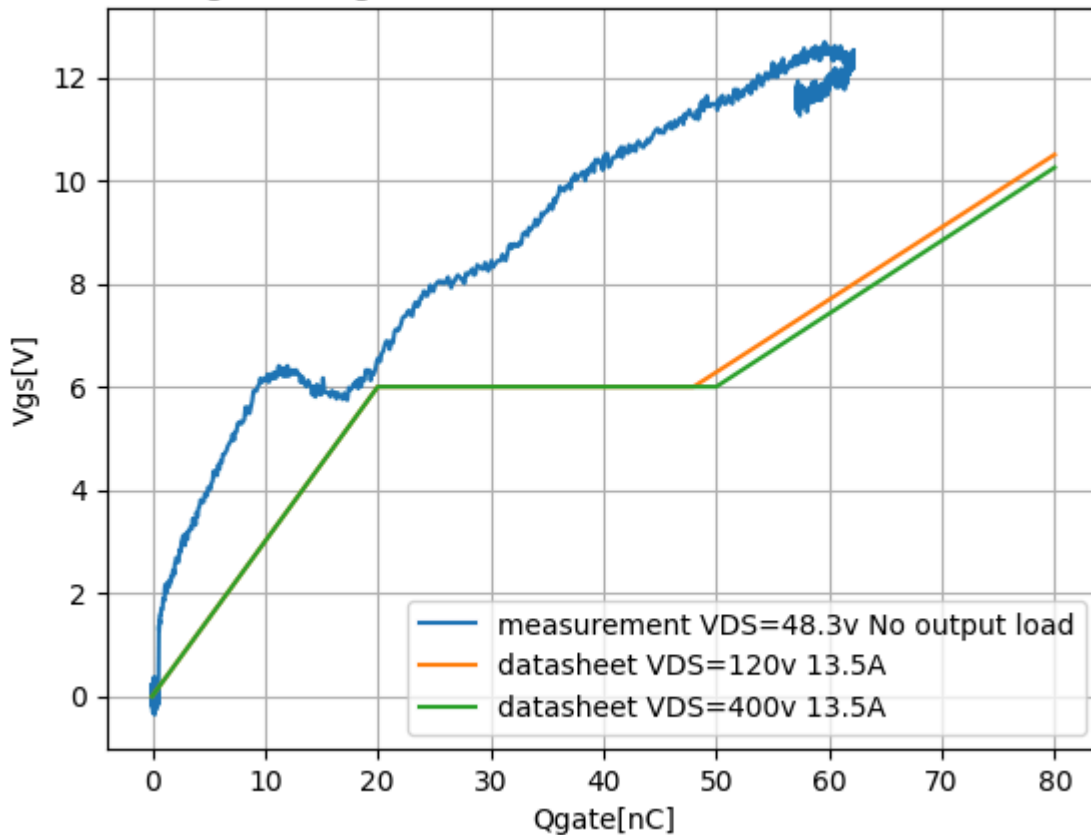
$$\begin{aligned} V_{DRV} &= 12 \text{ V} ; V_{gsTpeak} = 6 \text{ V} ; I_{gTpeak} = 0.730 \text{ A} ; R_{gTot} = \frac{V_{DRV} - V_{gsTpeak}}{I_{gTpeak}} = \frac{12 - 6}{0.730} = 8.219 \text{ } \Omega \end{aligned}$$

Theoretical value :

$$\begin{aligned} R_{gMosfet} &= 0.860 \text{ } \Omega ; R_{driver} = 5 \text{ } \Omega ; R_{ext} = 3 \text{ } \Omega ; R_{tot} = R_{gMosfet} + R_{driver} + R_{ext} = 0.860 + 5 + 3 = 8.860 \text{ } \Omega \end{aligned}$$

The results are very close

MOSFET: Miller plateau

Turn ON, gatecharge Mosfet IPW60R037CM8XKSA1, $V_{dd} = 11.35V$ 

The Miller plateau is not very pronounced because the SW-ON is in ZVS mode and at low voltage.

Secondary driver

Reminder of the secondary-side synchronous rectifier driver

the master ic : NCP43080DDR2G

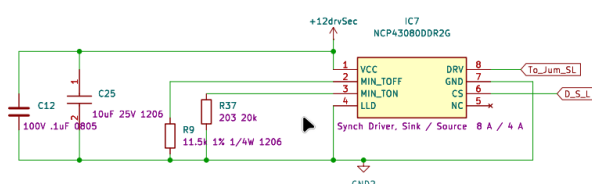


Figure 18 The driver NCP43080DDR2G

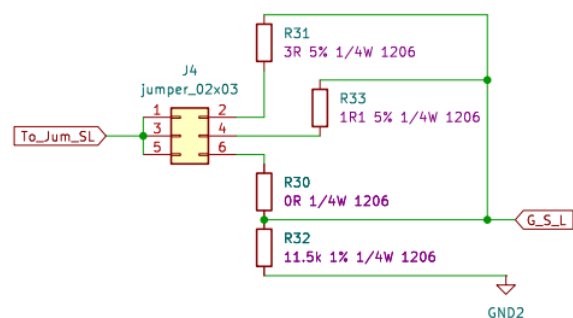


Figure 19 Jumper to shose R_g from 0, 1, 3 ohm

The controller presents a limitation, its conduction time depends heavily on the load; see the test below.

Test conditions:

- $V_{in} = 393.5 \text{ V}$
- $V_{out} = 48.11 \text{ V}$

Oscilloscope:

- CH1 : V_{ds} low side primary x 100
- CH2 : V_{gs} low side secondary: driver output
- CH4 : V_{ds} low side secondary x 10

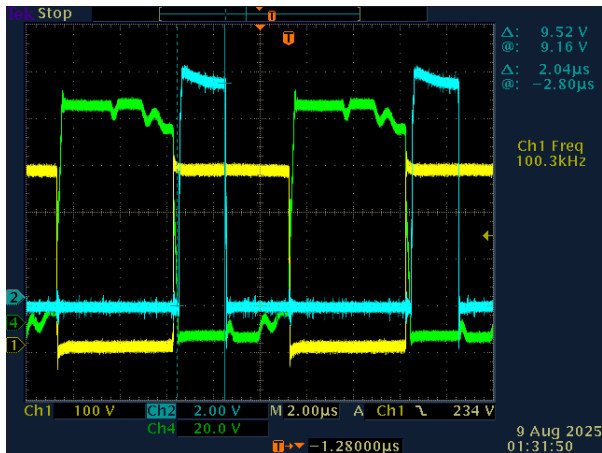


Figure 20 no load

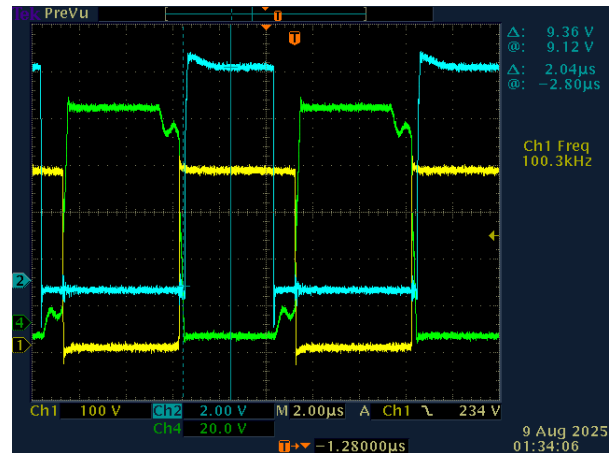


Figure 21 load1 52.1 ohm

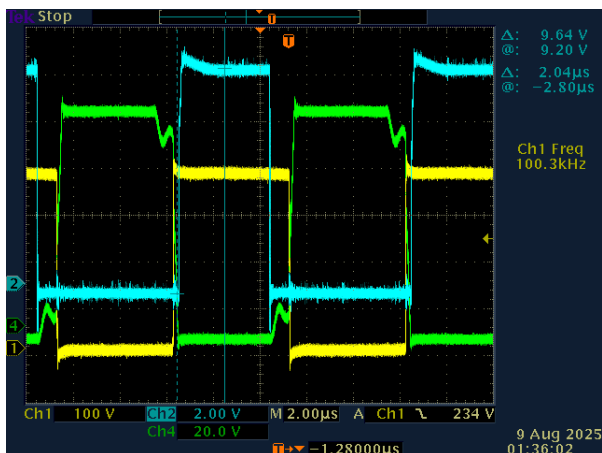


Figure 22 load2 36 ohm

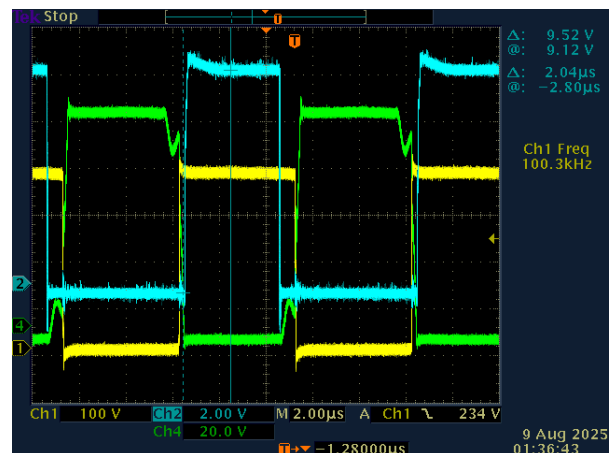


Figure 23 load3 3 24.5 ohm

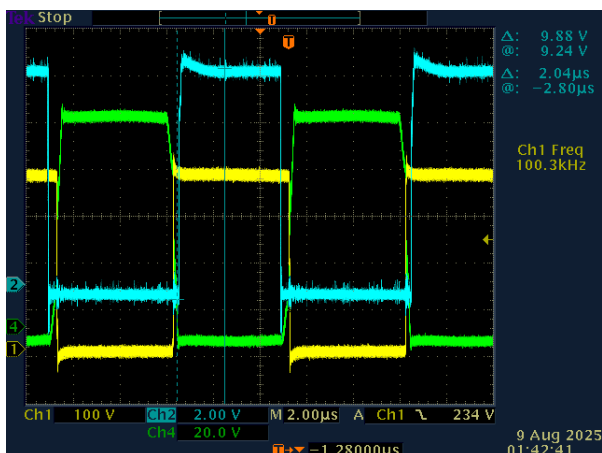


Figure 24 load4 12.8 ohm

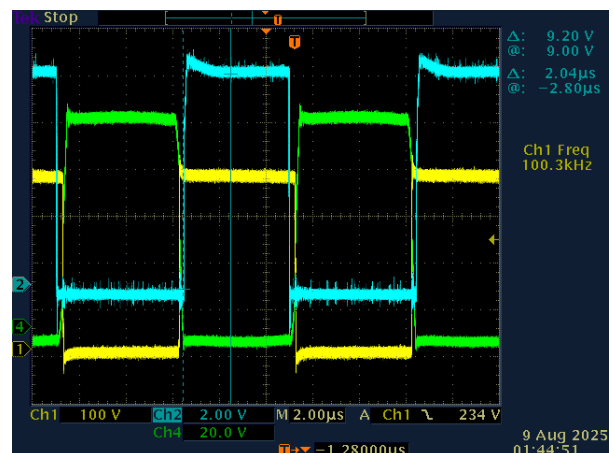


Figure 25 load5 9.2 ohm

Grouping of all test data :

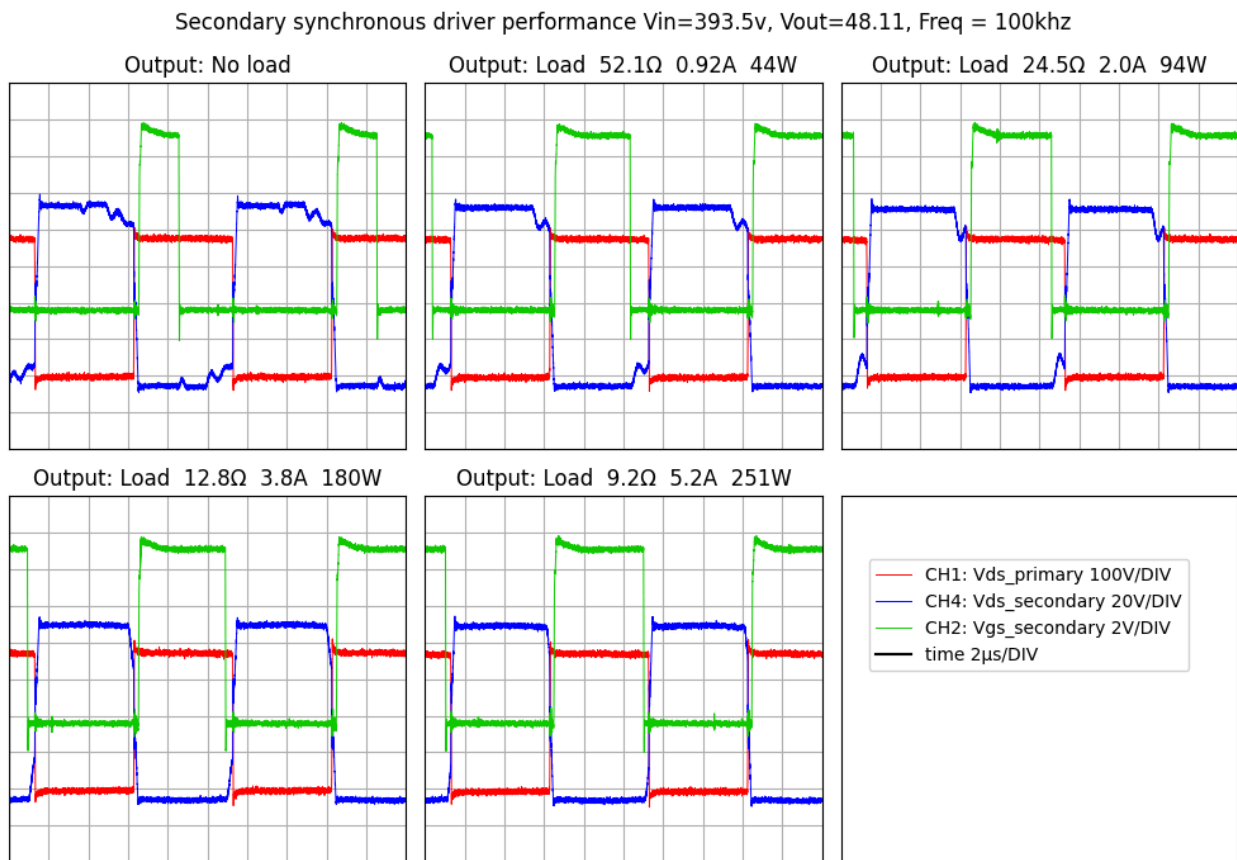


Figure 26 PreProcessing of the test results

Annexes

Carat of the big Lm

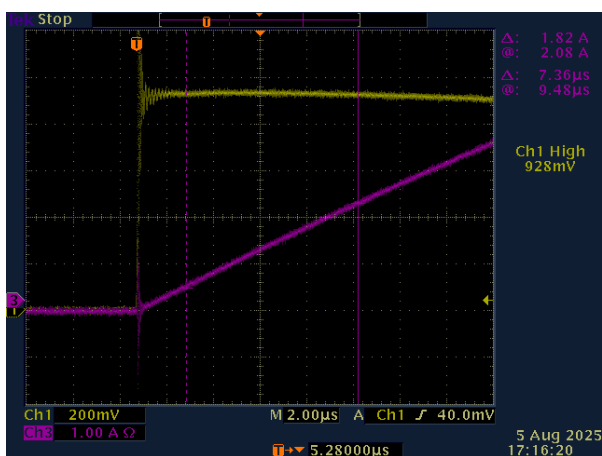


Figure 27 Current

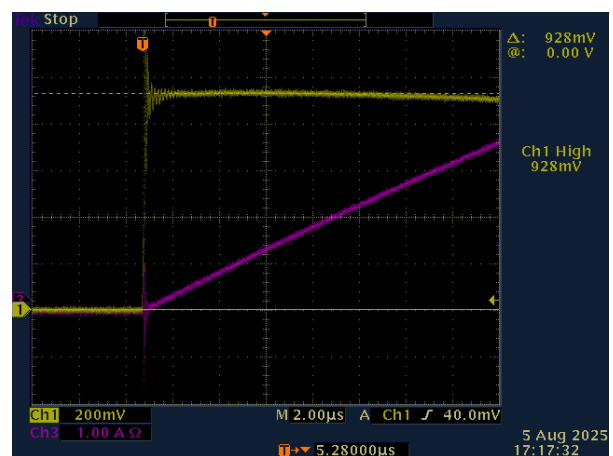


Figure 28 Voltage

We used the same double-pulse setup to characterize the large 16-turn inductor.

See the oscilloscope above (CH1 is and isolated x 100 prob)

$$\begin{aligned} N &= 16 \text{ turns} \\ \Delta I &= 1.820 \text{ A} \\ \Delta t &= 7.360 \text{ } \mu\text{s} \\ V_L &= 928 \text{ } \cdot 100 \text{ } \cdot 1 \text{ } \cdot 10^{-3} = 92.800 \text{ V} \end{aligned}$$

$$\text{Since } V_L = L \frac{di}{dt}$$

$$\begin{aligned} L &= V_L \cdot \left(\frac{\Delta t}{\Delta I} \right) = 92.800 \cdot \left(\frac{7.360}{1.820} \right) = 375.279 \text{ } \mu\text{H} \\ \text{AL} &= 1 \text{ } \cdot 10^3 \cdot \frac{L}{N^2} \\ &= 1 \text{ } \cdot 10^3 \cdot \frac{375.279}{(16)^2} = 1465.934 \text{ nH/turn}^2 \end{aligned}$$

Quick prototyping of rogowski coil

During this project, we experimented with several methods to produce a prototype of a Rogowski coil, The idea is to prototype a very simple Rogowski coil and perform the integration via signal preprocessing.

the produced signal is:

$$e = k \cdot \frac{di}{dt}$$

So a preprocessing integral is necessary and an empirical estimation of the coed k

Below some image of the prototyped coils:



Figure 29

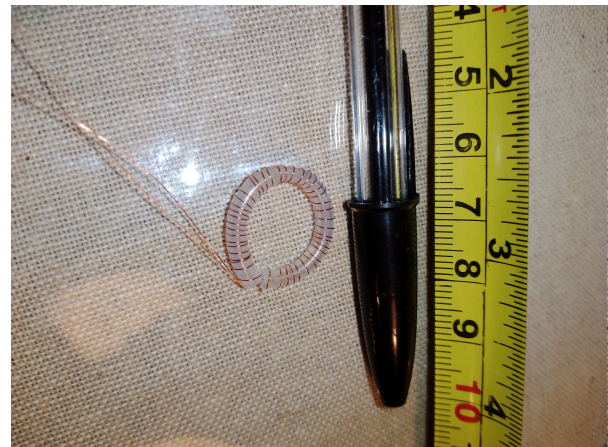


Figure 30

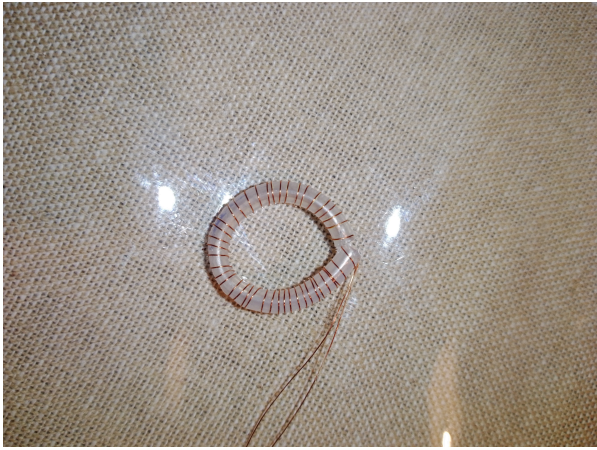


Figure 31

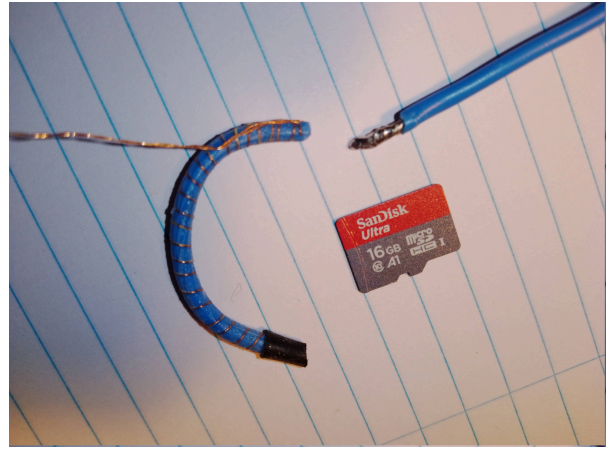


Figure 32

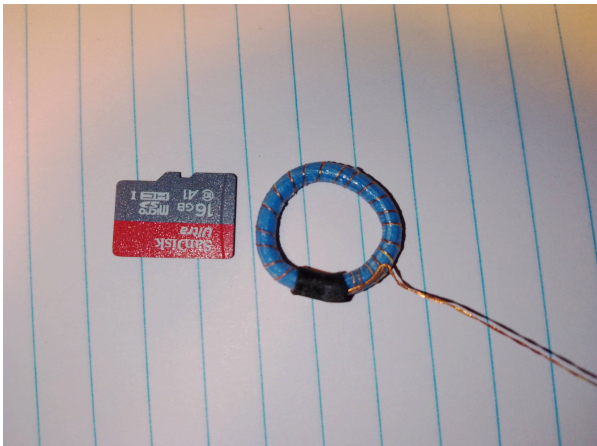


Figure 33



Figure 34

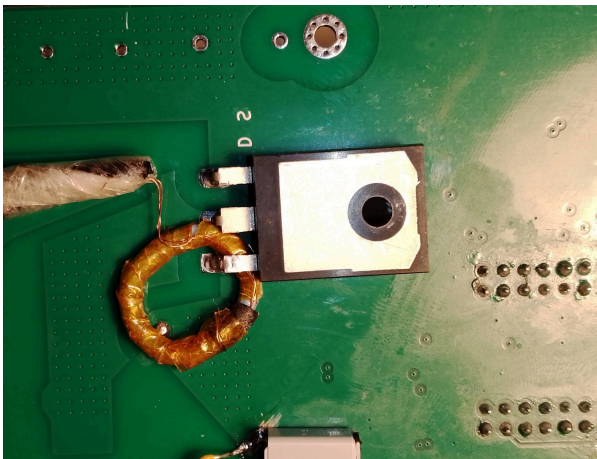


Figure 35

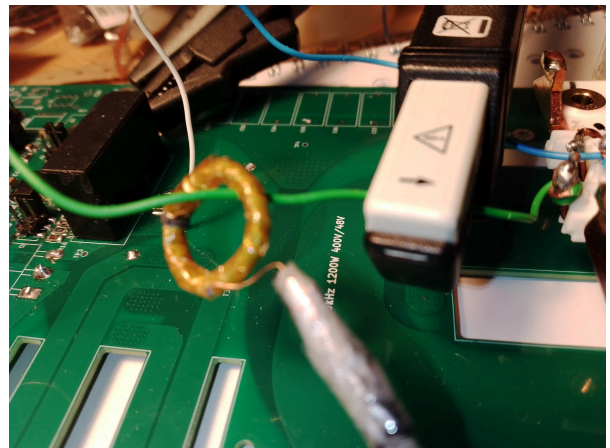


Figure 36

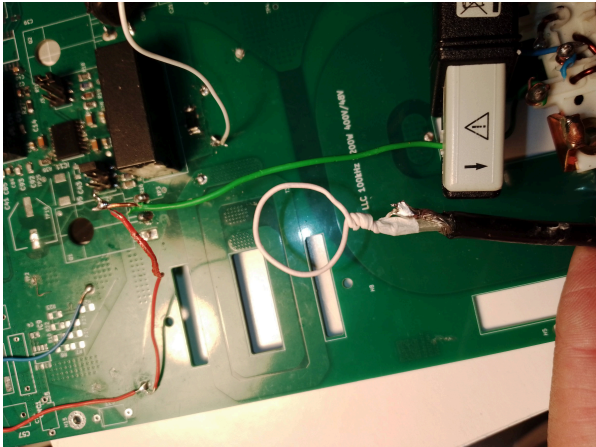


Figure 37

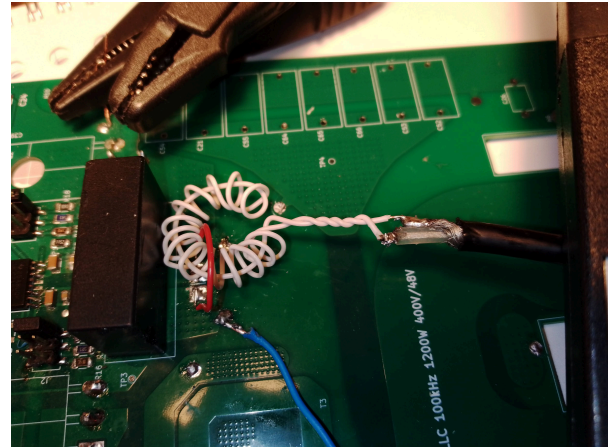


Figure 38

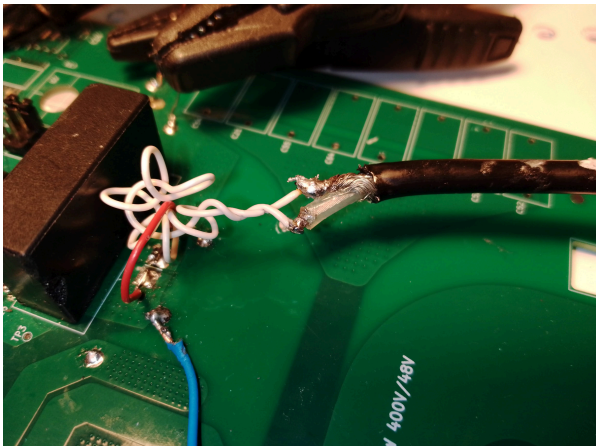


Figure 39

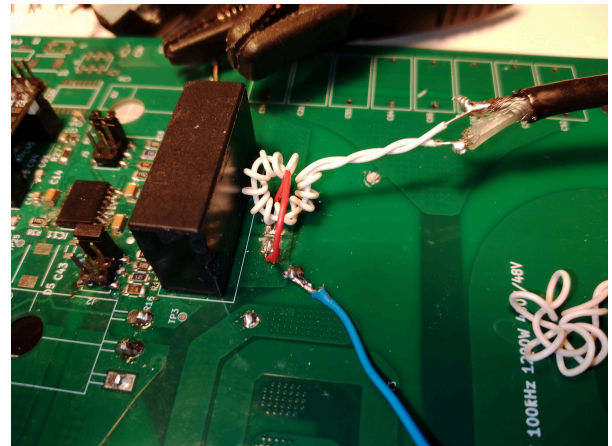


Figure 40

The preprocessing

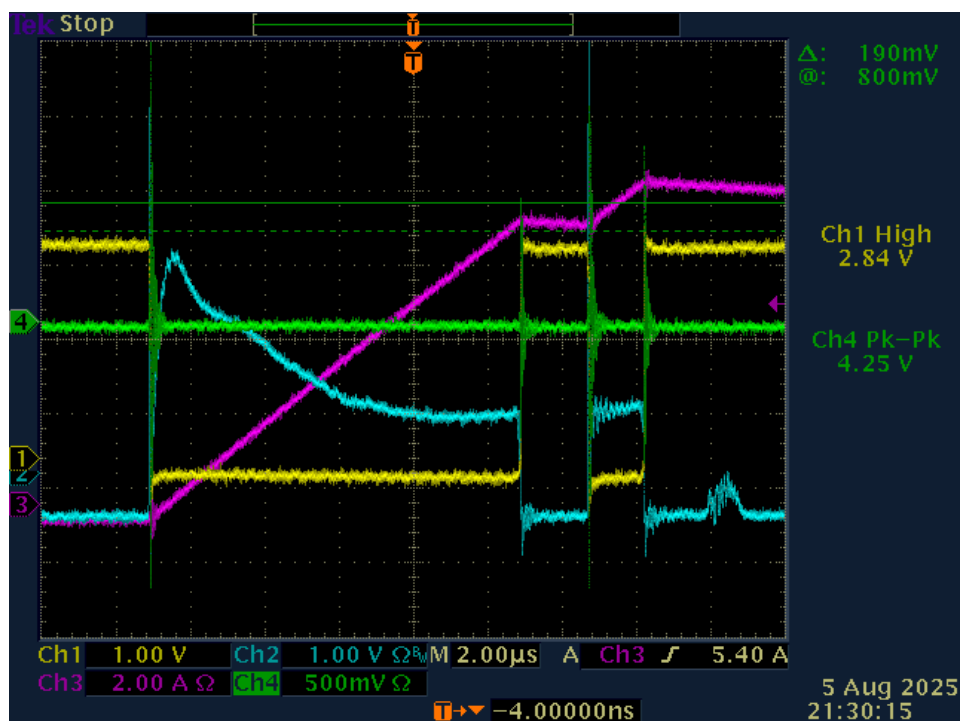
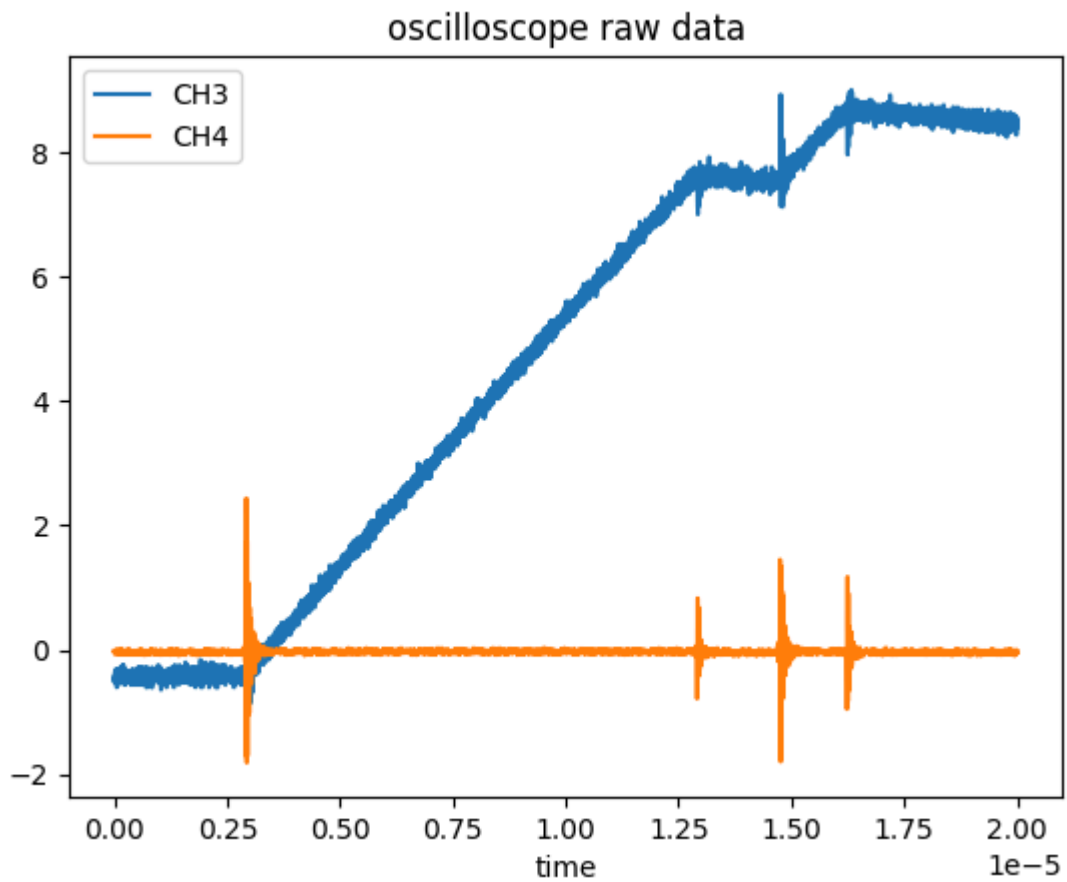
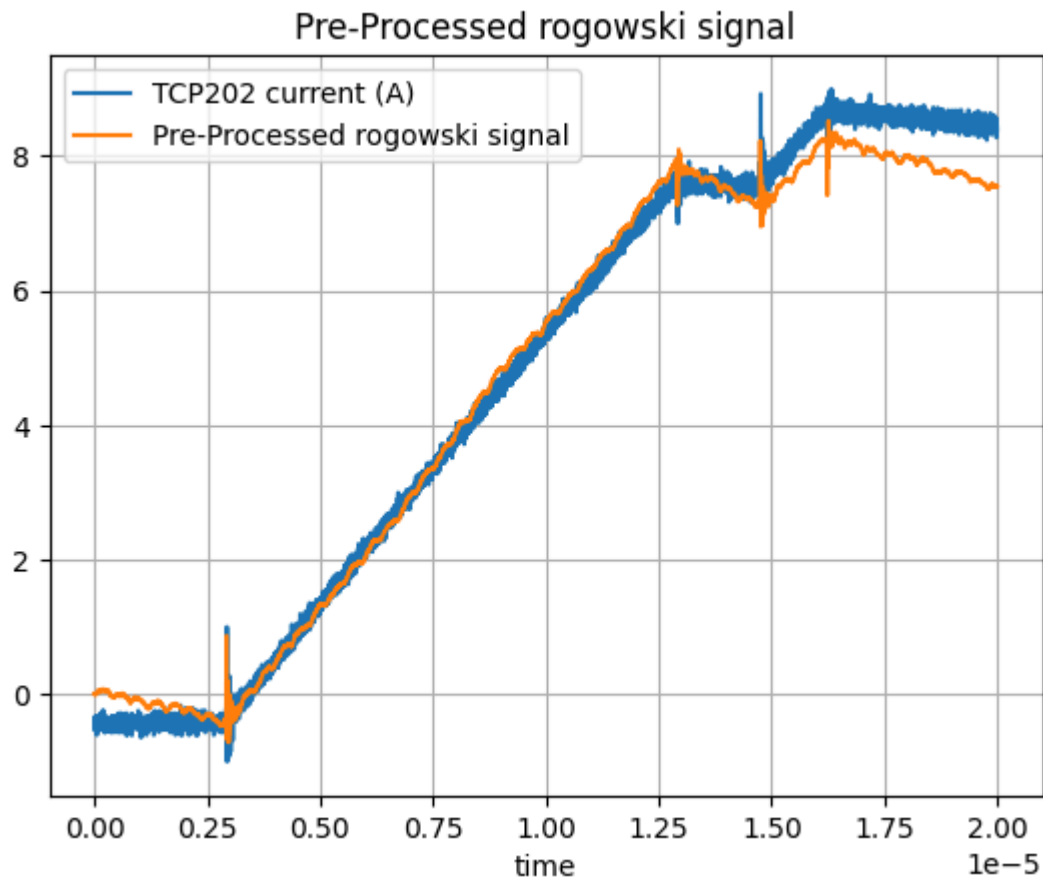


Figure 41 Example of measurement

- CH3 is TCP202 360 μ H current
- CH4 Rogowski coil voltage (terminal 50ohm)





Preprocessed rogowski signal is very good to detect fast change of the current but bad in low frequency area

Double pulse code STM32f103

Simple double pulse code

```
#include "stm32f1xx.h"

#define CPU_FREQ 72000000UL
#define PA9_SET (1u << 9)
#define PA9_RST (1u << (9 + 16))
#define PA9_BIT (1u << 9)

static inline void dwt_start(void) {
    CoreDebug->DEMCR |= CoreDebug_DEMCR_TRCENA_Msk;
    DWT->CYCCNT = 0;
    DWT->CTRL |= DWT_CTRL_CYCCNTENA_Msk;
}

void setup(void) {
    // Enable GPIOA clock
    RCC->APB2ENR |= RCC_APB2ENR_IOPAEN;

    // PA9 = General purpose push-pull output, 50 MHz (CRH bits [7:4])
    GPIOA->CRH &= ~(0xFu << 4);
    GPIOA->CRH |= (0x3u << 4); // MODE9=11 (50MHz), CNF9=00 (GP PP)
```

```

// Ensure USART1 doesn't grab PA9 (we don't enable it)
RCC->APB2ENR &= ~RCC_APB2ENR_USART1EN;

dwt_start();

// Quick self-check: force LOW and verify ODR reflects it
GPIOA->BSRR = PA9_RST;
if (GPIOA->ODR & PA9_BIT) { for (;;){ /* trap if config failed */ } }
}

void loop(void) {
    uint32_t ton1 = 2; // us
    uint32_t toff1 = 1; // us
    uint32_t ton2 = 1; // us
    uint32_t toff2 = 1; // us

    GPIOA->BSRR = PA9_SET;
    delayMicroseconds(ton1);

    GPIOA->BSRR = PA9_RST;
    delayMicroseconds(toff1);

    GPIOA->BSRR = PA9_SET;
    delayMicroseconds(ton2);

    GPIOA->BSRR = PA9_RST;
    delayMicroseconds(toff2);
}

```

Train of double pulse

```

#include "stm32f1xx.h"

// --- PA9 macros ---
#define PA9_SET    (1u << 9)
#define PA9_RST    (1u << (9 + 16))

// --- PB14 macros ---
#define PB14_SET   (1u << 14)
#define PB14_RST   (1u << (14 + 16))

#define _Ena_drv  PB10
#define _Dis_drv  PB12

void setup() {
    // Force PB12 HIGH in output data register BEFORE pinMode changes the mode
    // digitalWrite(_Dis_drv, HIGH);      // preload ODR = 1 (float high in
    // open-drain mode)
    digitalWrite(_Dis_drv, LOW);
    pinMode(_Dis_drv, OUTPUT_OPEN_DRAIN);
    digitalWrite(_Dis_drv, LOW); // redundancy
    digitalWrite(_Ena_drv, HIGH);
    pinMode(_Dis_drv, OUTPUT);
    digitalWrite(_Ena_drv, HIGH);
}

```



```

////////// PA9 & PB14 as Fast output //////////
// Enable GPIOA + GPIOB clocks
RCC->APB2ENR |= RCC_APB2ENR_IOPAEN | RCC_APB2ENR_IOPBEN;

// --- PA9 = 50 MHz push-pull output ---
GPIOA->CRH &= ~(0xF << 4); // clear config bits
GPIOA->CRH |= (0x3 << 4); // MODE9=11 (50MHz), CNF9=00
GPIOA->BSRR = PA9_RST; // start LOW

// --- PB14 = 50 MHz push-pull output ---
GPIOB->CRH &= ~(0xF << 24); // PB14 config bits
GPIOB->CRH |= (0x3 << 24); // MODE14=11, CNF14=00
GPIOB->BSRR = PB14_RST; // start LOW

}

```

```

__attribute__((always_inline))
static inline void delayNs(uint32_t tns)
{
    /*
    target => oscilo
    6000ns => 5986ns
    1000ns => 985.3ns
    */

    uint32_t iterations = (tns * 72) / (6.08*1000);

    for (uint32_t i = 0; i < iterations; i++) {
        __asm__ __volatile__("nop" ::: "memory");
    }
}

```

```
void loop() {
```

```

////////// ENABLE DRIVER
delayMicroseconds(10);
digitalWrite(_Dis_drv, HIGH); // redundancy
digitalWrite(_Ena_drv, LOW);
delayMicroseconds(10);
digitalWrite(_Ena_drv, HIGH);

////////// double pulse waveform on / off / on /off

uint32_t ton = 1000; // ns

```

```

uint32_t toff = 800; // ns
uint32_t N = 17; // n of pulses

for (int i = 0; i<N; i++){
    GPIOA->BSRR = PA9_SET;
    delayNs(ton);
    GPIOA->BSRR = PA9_RST;
    delayNs(toff);
}

while(1);

}

```

Miller effect: Simple explanation

Miller effect: Simple modelization

In this part, we propose a simple explanation of miller effect in the mosfet

From the **IPW60R037P7** [datasheet](#), we can found :

$$\begin{aligned} \mathrm{Q_{gs}} &= 27 \text{ nC} \\ \mathrm{Q_{gd}} &= 37 \text{ nC} \\ \mathrm{V_{GS}} &= 10 \text{ V} \\ \mathrm{V_{DD}} &= 400 \text{ V} \\ \mathrm{C_{CO}} &= 1599 \text{ pF (time related)} \\ V_{\text{gsPlateau}} &= 5.200 \text{ V} \end{aligned}$$

Calculating the average capacitors (actual capacitors depend on the voltage, but we will take this hypothesis)

$$\begin{aligned} \mathrm{C_{gd}} &= \frac{\mathrm{Q_{gd}}}{\mathrm{V_{DD}} - \mathrm{V_{GS}}} = \frac{37}{400 - 10} = 0.095 \text{ nF} \\ \mathrm{C_{gs}} &= \frac{\mathrm{Q_{gs}}}{V_{\text{gsPlateau}}} = \frac{27}{5.200} = 5.192 \text{ nF} \end{aligned}$$

- In the below simulation, we modulate the mosfet with ideal switch and capacitors
- We use a simple capacitor circuit to make integral of the gate current
- For C_{gs} , we will use a value of 8 nF to compensate for the non-linearity of this capacitor.

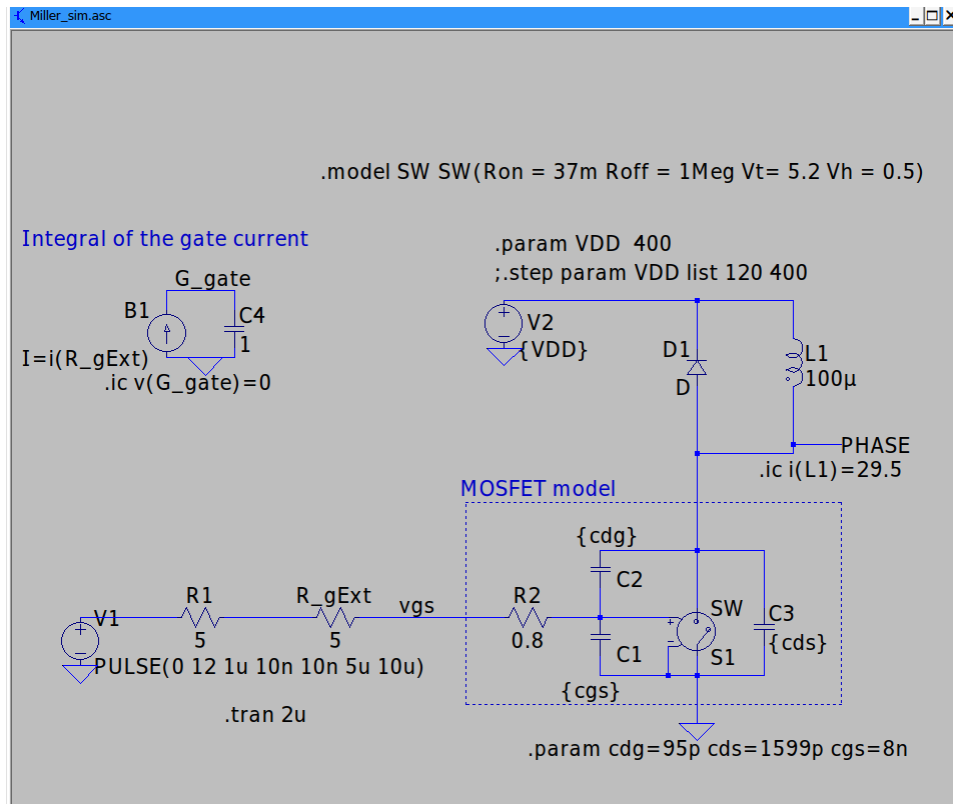


Figure 42 Simple LTSpice simulation

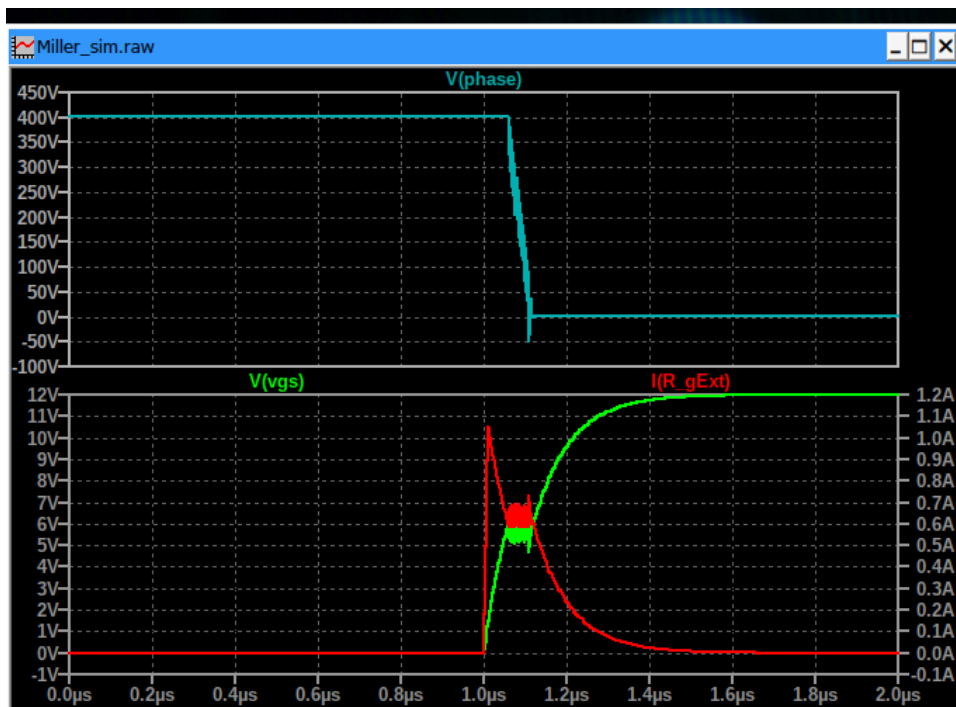


Figure 43 Simple LTSpice simulation: Time result

Result:

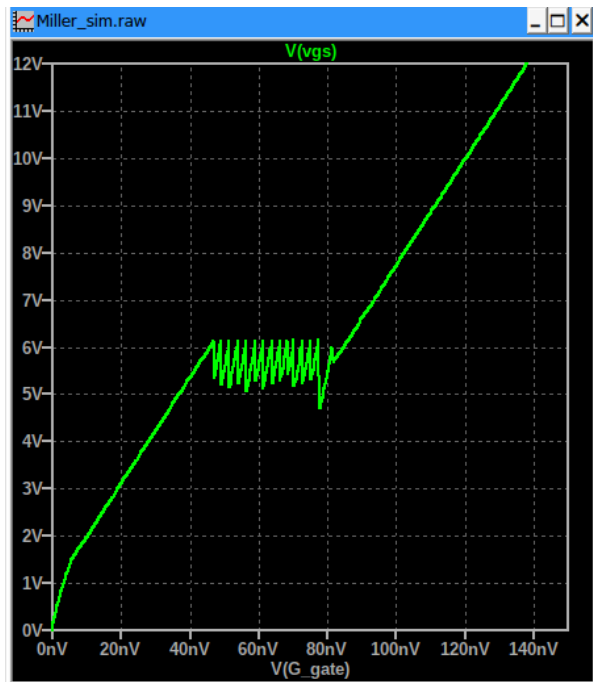


Figure 44 Simple LTSPice simulation: Result

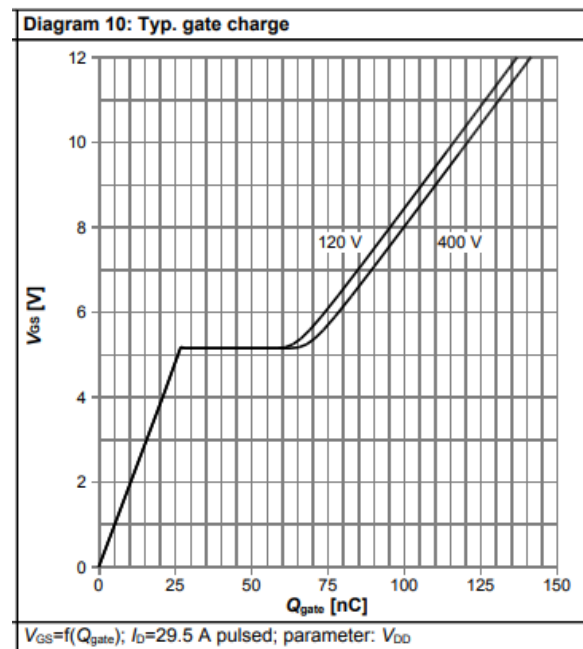


Figure 45 Datasheet

Using the very simple MOSFET model, we have already identified the overall signature of the Miller plateau:

- $\Delta Q = 65 - 25 = 40 \text{ nC}$ (datasheet)
- $\Delta Q = 34 \text{ nC}$ (Simulation)

Why an oscillation? Because the model is very simple: no inductance loop, no resistance, constant capacitor, ideal diode...

Miller effect cross-conduction problem

One of the challenges associated with the Miller effect concerns the turn-off phase:

To illustrate this problem, shown below is a simple simulation model using a half-bridge and an LC branch to achieve ZVS.

- The gate resistance is identical for both SW-on / SW-off.
- It can be observed that the drain current (in purple) make a 200A spike (caused by a short circuit between the high/low side MOSFETS).

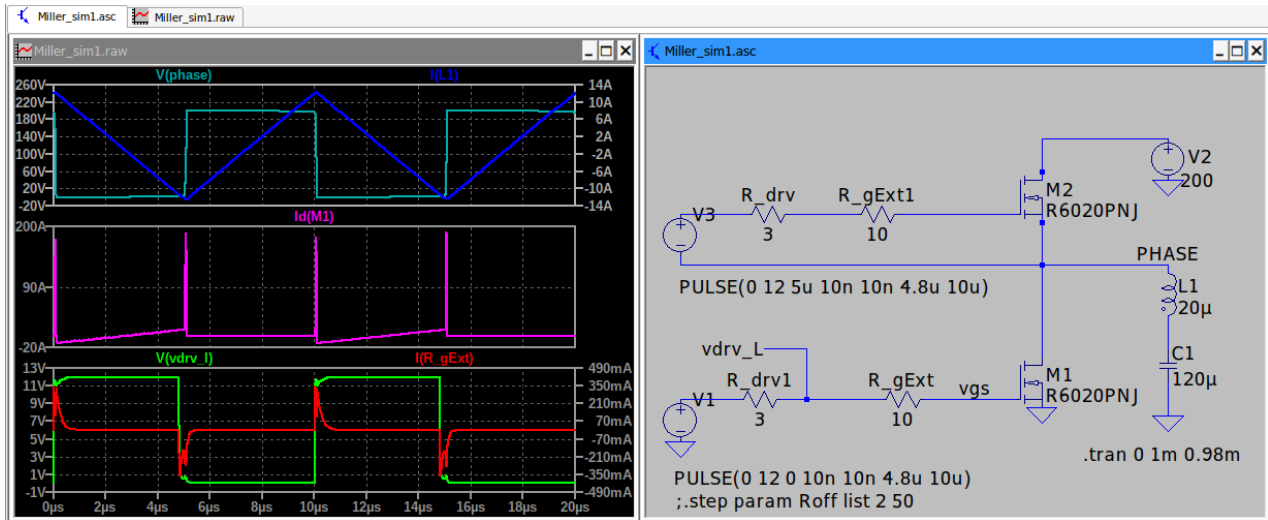


Figure 46 Simple LTSpice simulation: NO ROFF branch, dead time =200ns

By adding a resistor R_{off} , the discharge time is fast and no short circuit occurs

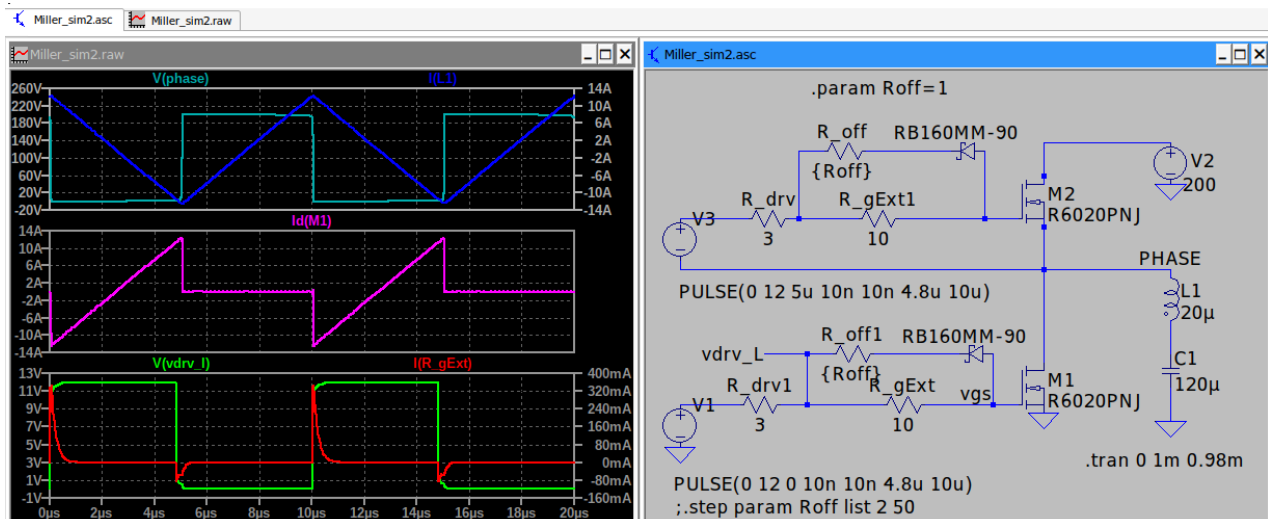


Figure 47 Simple LTSpice simulation: with ROFF branch, dead time =200ns